Simple Impedance Response Formulas for the Dispersive Interaction Rates in the Effective Hamiltonians of Low Anharmonicity Superconducting Qubits

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***Abstract*— For superconducting quantum processors consisting of low anharmonicity qubits such as transmons, we give a complete microwave description of the system in the qubit subspace. We assume that the qubits are dispersively coupled to a distributed microwave structure such that the detunings of the qubits from the internal modes of the microwave structure are stronger than their couplings. We define "qubit ports" across the terminals of the Josephson junctions and "drive ports" where transmission lines carrying drive signals reach the chip and we obtain the multiport impedance response of the linear passive part of the system between the ports. We then relate interaction parameters in between qubits and between the qubits and the environment to the entries of this multiport impedance function; in particular, we show that the exchange coupling rate J between qubits is related in a simple way to the off-diagonal entry connecting the qubit ports. Similarly, we relate couplings of the qubits to voltage drives and lossy environment to the entries connecting the qubits and the drive ports. Our treatment takes into account all the modes (possibly infinite) that might be present in the distributed electromagnetic structure and provides an efficient method for the modeling and analysis of the circuits.**

***Index Terms*— Josephson junctions, microwave circuits, quantum Hamiltonians, superconducting qubits.**

I. Introduction

SUPERCONDUCTING circuits are a promising platform for the realization of quantum computers. Operated at microwave frequencies they include Josephson junctions for the nonlinearity needed to obtain qubit modes without introducing significant loss. Coherence times of the superconducting qubits have been improved by several orders of magnitude in the last two decades, and the Transmon qubit [1], [2] (and its several variations [3], [4]) has now become the superconducting qubit of choice in many groups around the world due to its simplicity of design and its superior coherence. Fidelities of the single-qubit gates are now routinely below [5], [6] and those of the two-qubit gates are at the fault-tolerance threshold levels required by the surface code [5], [7], [8]. The challenge now is to scale the circuits up while maintaining and improving further the qubit coherence times and gate fidelities [9]. Many important engineering problems, however, arise in the design of larger multiqubit devices, such as signal crosstalk and qubit–qubit crosstalk, which show the need for better models/tools to understand and improve the operation of the superconducting quantum processors.

Several methods have been used to model and study the physics of superconducting qubit circuits. The Jaynes–Cummings model [10] originally introduced in quantum optics has routinely been applied to the study of the so-called circuit-QED architecture [11], [12], in which superconducting qubits are coupled to readout resonators for their control and readout, and two-qubit gate operations are mediated by the bus resonators. Readout and bus resonators are typically designed to be detuned away from the qubits to operate in the so-called dispersive regime. In that regime, one can eliminate the resonators up to desired order in the bare qubit-resonator couplings and get an effective description of the system in the qubit subspace. However, calculation of the dispersive quantities such as the exchange coupling or Purcell decay rates [13] of the qubits with the single mode Jaynes–Cummings model showed significant discrepancy with the experimental measurements and attempts to include higher harmonics of the resonators with multimode extensions of the Jaynes-Cummings model failed due to divergence issues [14]. Reference [15] showed the convergence of the Lamb shift in the specific case of a Josephson junction atom coupled to a multimode resonator in the Rabi model. More recently, [16] studied the convergence of the bare couplings between the superconducting qubits and multimode resonators in various general coupling configurations.

Combination of lumped element circuit quantization methods [17]–[19] with classical circuit synthesis techniques [20]–[22] resulted in "blackbox quantization" methods [25]–[27], which allowed extraction of the parameters in the quantum Hamiltonian models of the superconducting circuits from the electromagnetic finite-element simulations. The simulations correspond to the linear passive part of the circuits, which is usually a distributed microwave structure as seen looking into the ports defined across the Josephson junctions.

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Although such an approach allows an accurate treatment of very general structures consisting possibly of multiple microwave modes, the simulation of large multiqubit devices might quickly become computationally demanding.

Following a similar approach, we show here that for superconducting processors consisting of low anharmonicity qubits, such as transmons, the dispersive interaction parameters, such as exchange coupling and Purcell decay rates of the qubits and their coupling to the voltage drives, are related in a simple way to the microwave impedance response functions as seen at the "qubit ports" and "drive ports." This reduces a large portion of the design of multiqubit superconducting devices into a classical microwave engineering problem (up to the assumptions and approximations we are making here) and allows one to avoid any numerical multimode block diagonalization or fitting of electromagnetic finite-element simulations over a range of frequencies which are both expensive if not prohibitive computational procedures.

We propose the following effective Hamiltonian to describe a multiqubit superconducting device operated in the dispersive regime and consisting of low anharmonicity qubits coupled to each other and to the external world through a linear passive distributed microwave structure (assuming that the detunings of the internal modes of the microwave structure from the qubits are stronger than their couplings):

$$\hat{\mathcal{H}}/\hbar = \hat{H}_Q + \hat{H}_\chi + \hat{H}_R \tag{1}$$

where $\hat{\mathcal{H}}$ is the block diagonal with $\hat{H}_Q$ and $\hat{H}_R$ being the subblocks corresponding to the qubit and resonator subspaces, respectively, and $\hat{H}_\chi$ is the dispersive interaction term, the only interaction left between the qubit and resonator modes. $\hat{H}_Q$ consists of three terms

$$\hat{H}_Q = \hat{H}_Q^D + \hat{H}_Q^J + \hat{H}_Q^V \tag{2}$$

with $\hat{H}_Q^D$ being the diagonal part of $\hat{H}_Q$

$$\hat{H}_Q^D = \sum_{i=1}^{N} \omega_i \hat{b}_i^\dagger \hat{b}_i + \frac{\delta_i}{2} \hat{b}_i^\dagger \hat{b}_i \left(\hat{b}_i^\dagger \hat{b}_i - 1\right) \tag{3}$$

where we have N qubit modes represented as Duffing oscillators in the harmonic oscillator basis; $\hat{b}_i^{(\dagger)}$ is the annihilation (creation) operator of the qubit mode i of frequency ω_i given by $\omega_i = \omega_{J_i} - (E_C^{(i)}/\hbar/1 - E_C^{(i)}/(\hbar\omega_{J_i}))$ with $\omega_{J_i} = 1/\sqrt{L_{J_i} C_i}$, L_{J_i} being the bare junction inductance, and $E_C^{(i)} = (e^2/2C_i)$ being the charging energy of the qubit i of total shunt capacitance C_i and anharmonicity δ_i given by $\delta_i = -E_C^{(i)}(\omega_{J_i}/\omega_i)^2$. Such an approximate description on the harmonic basis is valid for qubits with low anharmonicity $\delta_i \ll \omega_i$ such as transmons.

$\hat{H}_Q^J$ in (2) gives the exchange coupling between the qubits

$$\hat{H}_Q^J = \sum_{i,j} J_{ij} \left(\hat{b}_i \hat{b}_j^\dagger + \hat{b}_i^\dagger \hat{b}_j\right). \tag{4}$$

As we show in the following, the exchange coupling rate J_{ij} between qubit modes i and j in such an effective description is a simple function of the impedance response defined between the "qubit ports":

$$J_{ij} = -\frac{1}{4}\sqrt{\frac{\omega_i \omega_j}{L_i L_j}} \mathrm{Im}\left[\frac{Z_{ij}(\omega_i)}{\omega_i} + \frac{Z_{ij}(\omega_j)}{\omega_j}\right] \tag{5}$$

where L_i and L_j are the "qubit inductances" corresponding to the qubits i and j, respectively, related to the bare junction inductances L_{J_i} by $L_i = L_{J_i}/(1 - 2E_C^{(i)}/\hbar\omega_i)$ such that $\omega_i = 1/\sqrt{L_i C_i}$. $Z_{ij}(\omega)$ is the (i, j)th entry of the multiport impedance matrix $\mathbf{Z}(\omega)$ connecting the ith qubit's port to the jth qubit's port. Qubit ports are defined between the terminals of the Josephson junctions, i.e., port voltages are voltages developed across and the port currents are the currents flowing through the Josephson junctions (see also Appendix D for how to define qubit ports as lumped ports in electromagnetic simulators). The multiport impedance matrix $\mathbf{Z}(\omega)$ is to be computed between the qubit ports with Josephson junctions removed. $\mathbf{Z}(\omega)$ then gives the response of the linear part of the circuit seen by looking into the qubit ports; in particular, $Z_{ij}(\omega)$ is the voltage developed across the ith qubit's port while a current of unit magnitude and frequency ω is driving the jth qubit's port while all other qubit ports left open. We note here that the formula in (5) holds in the case of a distributed microwave structure consisting of multiple internal modes (possibly infinite) coupling the qubits.

$\hat{H}_Q^V$ in (2) is the term giving the coupling of the qubits to the voltage drives

$$\hat{H}_Q^V = \sum_{i=1}^{N} \sum_{d=1}^{N_D} \varepsilon_{id} \left(\hat{b}_i - \hat{b}_i^\dagger\right) V_d \tag{6}$$

where we have N_D voltage sources and V_d is the voltage source driving the dth drive line for $1 \le d \le N_D$ (assuming there are a total of N_D lines driving the system as shown in Appendix B). ε_{id} is the matrix entry giving the coupling of the qubit i to the voltage source V_d. In Section B, we show that (under the assumption that no off-chip crosstalk is happening between the drive lines)

$$\varepsilon_{id} = \sqrt{\frac{\omega_i}{2\hbar L_i}} \mathrm{Im}[Z_{i,p(d)}(\omega_i)] \frac{e^{i\theta_d} C_{p(d)}}{\sqrt{1 + \omega_d^2 Z_0^2 C_{p(d)}^2}} \tag{7}$$

where $\theta_d = (\pi/2) - \arctan(\omega_d Z_0 C_{p(d)})$ and $Z_{i,p(d)}(\omega_i)$ is the entry of the multiport impedance matrix connecting the drive port [with port index $p(d)$] corresponding to the voltage source V_d (for the definition of drive ports, see Section IV-A and Appendix D) to the qubit port i evaluated at the frequency ω_i of qubit i; ω_d is the frequency of the voltage source V_d (assuming a single-tone sinusoidal signal), Z_0 is the characteristic impedance of the drive lines which is typically $Z_0 = 50\ \Omega$, and $C_{p(d)}$ is the shunting capacitance of the drive port corresponding to the voltage source V_d . Since the drive ports are defined where the drive lines reach the chip, the factor $\mathrm{Im}[Z_{i,p(d)}(\omega_i)]$ in (7) gives the classical crosstalk happening at the transition region where the lines land onto the chip or on the chip. We also propose below the following expression in units of dB as a measure of the classical crosstalk assuming

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similar values for qubit parameters in (7):

$$X_{ij} = 20\log_{10}\left(\left|\frac{\mathrm{Im}[Z_{i,d(j)}(\omega_i)]}{\mathrm{Im}[Z_{j,d(j)}(\omega_j)]}\right|\right) \tag{8}$$

where $d(j)$ is the port index of the drive of the qubit j. X_{ij} is the voltage crosstalk in dB seen by qubit i while driving qubit j.

Similar to $\hat{H}_Q$ in (2), $\hat{H}_R$ in (1) consists of three terms

$$\hat{H}_R = \hat{H}_R^D + \hat{H}_R^J + \hat{H}_R^V \tag{9}$$

with $\hat{H}_R^D$ being the diagonal part of $\hat{H}_R$

$$\hat{H}_R^D = \sum_{k=1}^{M} \omega_{R_k}\hat{a}_k^\dagger\hat{a}_k + \frac{\chi_{kk}^{(R)}}{2}\hat{a}_k^\dagger\hat{a}_k\left(\hat{a}_k^\dagger\hat{a}_k - 1\right) \tag{10}$$

where we have M resonator modes with $\hat{a}_k^{(\dagger)}$ being the annihilation (creation) operator of the resonator mode k with frequency ω_{R_k} and anharmonicity (or self-Kerr) $\chi_{kk}^{(R)}$ (we will be using the terms "resonator mode" and "internal mode" interchangibly in the following to refer to the microwave modes of the distributed linear passive structure the qubits are connected to). Similar to qubit modes, the exchange interactions in between resonator modes are given by the term $\hat{H}_R^J$ in (9)

$$\hat{H}_R^J = \sum_{k,k'} J_{kk'}\left(\hat{a}_k\hat{a}_{k'}^\dagger + \hat{a}_k^\dagger\hat{a}_{k'}\right) \tag{11}$$

where $J_{kk'}^{(R)}$ is the exchange coupling rate between resonator modes k and k' mediated by the qubits. $\hat{H}_R^V$ in (9) gives the coupling of the internal modes to the voltage sources

$$\hat{H}_R^V = \sum_{k=1}^{M}\sum_{d=1}^{N_D} \varepsilon_{kd}\left(\hat{a}_k - \hat{a}_k^\dagger\right)V_d. \tag{12}$$

The only remaining interaction between the qubit and internal modes is given by the dispersive interaction term $\hat{H}_\chi$ in (1)

$$\hat{H}_\chi = \sum_{i=1}^{N}\sum_{k=1}^{M} \chi_{ik}\hat{b}_i^\dagger\hat{b}_i\hat{a}_k^\dagger\hat{a}_k \tag{13}$$

where χ_{ik} is the dispersive energy shift between the qubit mode i and the resonator mode k. The resonance frequency ω_{R_k} of the resonator k gets the dispersive shift χ_{ik} depending on the state of the qubit i. We calculate χ_{ik} in Section VI similar to what has been done in [25] by including up to sixth-order nonlinear terms in the junction potentials

$$\chi_{ik} = 8\delta_i\left(1 - \frac{6E_C^{(i)}}{\omega_i}\right)\left(\frac{g_{ik}\omega_{R_k}}{\omega_{R_k}^2 - \omega_i^2}\right)^2 \tag{14}$$

where g_{ik} is the bare coupling rate between the qubit mode i and the resonator mode k given in (20).

We assume that the losses in the system are small; in particular, we neglect any internal loss. Hence, $\mathrm{Im}[\mathbf{Z}(\omega)]$ describes the lossless part of the system to a very good approximation. In Section V, we describe how to include the effect of external losses due to the coupling to drive lines by computing Purcell rates for the qubit modes. We show that the Purcell loss rate $(1/T_1^{i,d})$ of qubit i due to the drive line d is

$$\frac{1}{T_1^{i,d}} = \frac{2}{L_i}\mathrm{Im}[Z_{i,p(d)}(\omega_i)]^2\frac{\omega_i^2 Z_0 C_{p(d)}^2}{1 + \omega_i^2 Z_0^2 C_{p(d)}^2}. \tag{15}$$

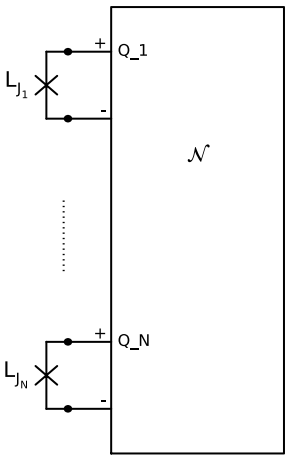


Fig. 1. Qubit ports labeled by Q_1 to Q_N are defined across the terminals (black dots) of the Josephson junctions. The blackbox $\mathcal{N}$ is an N-port linear passive network having the multiport impedance $\mathbf{Z}$ looking into the qubit ports with the partial-fraction expansion given in (16).

We note here that all the dispersive rates of qubit–qubit interactions and of interactions of qubits with the external electronics are functionals of the multiport impedance function $\mathbf{Z}(\omega)$ and bare junction inductances L_{J_i} since the shunting capacitances C_i of the qubit ports are related to the residue $\mathbf{A}_0$ of $\mathbf{Z}(\omega)$ at dc as given in (65) (the same argument applies to the shunt capacitances $C_{p(d)}$ of the drive ports) and the qubit frequencies ω_i and anharmonicities δ_i are the functions of qubit shunt capacitances and bare junction inductances. We also note that the terms of the form $\chi_{iikk'}\hat{b}_i^\dagger\hat{b}_i\hat{a}_k^\dagger\hat{a}_{k'}$ that are usually dropped by rotating wave approximation might be comparable to other terms in (1) if the frequencies ω_{R_k} and $\omega_{R_{k'}}$ of resonators k and k' are not detuned enough.

II. Derivation of the Formula for the Exchange Coupling Rates Between the Qubits

Assuming we have N Josephson junctions in the circuit, we define the $N \times N$ multiport impedance matrix $\mathbf{Z}$ seen looking into qubit ports defined across the junction terminals, as shown in Fig. 1. ($\mathbf{Z}$ has to be evaluated without shunting the qubit ports by Josephson junctions.) Neglecting all the losses, we can write the following partial fraction expansion for the imaginary part of $\mathbf{Z}(\omega)$ as a function of the frequency variable ω [22]:

$$\mathrm{Im}[\mathbf{Z}(\omega)] = -\frac{\mathbf{A}_0}{\omega} + \sum_{k=1}^{M}\frac{\mathbf{A}_k\omega}{\omega_{R_k}^2 - \omega^2} + \mathbf{A}_\infty\omega \tag{16}$$

where ω_{R_k} values are the frequencies of the internal modes corresponding to readout and bus resonators and $\mathbf{A}_k$ values are rank-1 (although [22] considers general full-rank matrices which would correspond to having multiple degenerate internal modes, we argue that, in real physical systems, small couplings will remove any such degeneracy) real symmetric positive semi-definite $N \times N$ matrices for $1 \le k \le M$. Although we have truncated the part corresponding to internal modes to M terms as we will see in the following, the formula in (5) stays valid in the limit of an infinite number of modes $M \to \infty$ (more generally, one can think of the multiport impedance

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expansion in (16) as being corresponding to any distributed electromagnetic structure seen by the junctions).

The impedance matrix $\mathbf{Z}$ needs to be *lossless positive real* (LPR) to have the partial-fraction expansion given in (16). Reference [22, Th. 5-1] gives the *positive real* (PR) test by listing necessary and sufficient conditions for a matrix $\mathbf{Z}(s)$ to be PR (where $s = \sigma + j\omega$ is the complex frequency variable). An $N \times N$ matrix $\mathbf{Z}(s)$ is PR if and only if it is really rational, has no poles in the right-half plane, has simple purely imaginary poles, its residues corresponding to the purely imaginary poles are Hermitian and positive semidefinite, and its Hermitian part evaluated on the imaginary axis is positive semidefinite. By [22, eqs. (3)–(5)], a PR matrix $\mathbf{Z}$ is lossless or LPR if it satisfies $\mathbf{Z}(s) = -\mathbf{Z}^T(-s)$.

The PR property was introduced by Brune [21] as a set of necessary and sufficient conditions for a function to correspond to the single-port impedance (driving-point impedance) of a finite passive network consisting of lumped elements. Given a PR function, Brune [21] also discovered an algorithm for the synthesis of a network (consisting of resistors, capacitors, inductors, and ideal transformers) admitting this function as its driving-point impedance. PR property is defined for finite networks by the real-rational condition. Reference [23] describes how to synthesize infinite networks consisting of passive lumped circuit elements corresponding to transcendental impedance functions of distributed electromagnetic structures such as finite-length transmission lines by showing PR property of each term in their partial-fraction expansions. For more general structures, one can perform an electromagnetic simulation to obtain the impedance response of the system over a finite frequency band, and using tools, such as vector fitting [24], fits the response to a finite pole-residue model as we have in (16). Vector fitting ensures the passivity and the stability of the model and one can increase the number of poles until the required accuracy is reached. We also note here that the LPR property described earlier is an extension of Foster's reactance theorem [20] to the multiport case.

Starting with the expansion in (16), we can synthesize a lossless multiport lumped element circuit [22] corresponding to the multiport blackbox network $\mathcal{N}$ in Fig. 1, as shown in Appendix A. Using the method described in [19], we can identify the degrees of freedom in this circuit and derive the following Hamiltonian (see Appendix A):

$$\mathcal{H} = \frac{1}{2}\mathbf{Q}^T\mathbf{C}^{-1}\mathbf{Q} + \frac{1}{2}\mathbf{\Phi}^T\mathbf{M}_0\mathbf{\Phi} - \sum_{i=1}^{N} E_{J_i}\cos(\varphi_{J_i}) \quad (17)$$

where $\mathbf{\Phi} = (\Phi_{J_1}, \ldots, \Phi_{J_N}, \Phi_{R_1}, \ldots, \Phi_{R_M})^T$ being the flux coordinate vector. φ_{J_i} is the phase of the junction i related to the flux across it by the Josephson relation $\Phi_{J_i} = (\Phi_0/2\pi)\varphi_{J_i}$, for $1 \le i \le N$. Φ_{R_k} is the flux across the inductor of the internal mode k, $1 \le k \le M$. E_{J_i} is the Josephson energy of junction i related to its inductance L_{J_i} by $E_{J_i} = (\Phi_0/2\pi)^2(1/L_{J_i})$. The capacitance matrix $\mathbf{C}$ is given by

$$\mathbf{C} = \begin{pmatrix} \mathbf{C}_0 & -\mathbf{C}_0\mathbf{R}^T \\ -\mathbf{R}\mathbf{C}_0 & \mathbf{1}_{M\times M} + \mathbf{R}\mathbf{C}_0\mathbf{R}^T \end{pmatrix} \quad (18)$$

where $\mathbf{C}_0$ is the diagonal with entries $(C_1, \ldots, C_N)$, and C_i being the total capacitance shunting the junction i. This is a valid physical assumption since it corresponds to having no direct electrostatic dipole–dipole interaction between junction terminals. Such an assumption will keep our discussion simple although the case of nondiagonal $\mathbf{C}_0$ will not change any of the results. In such a case, one can treat the nondiagonal part of $\mathbf{C}_0$ at frequency ϵ like the other terms at finite frequencies ω_{R_k} in the impedance expansion in (16) and apply the Schrieffer–Wolff transformation as described in the following in the limit of $\epsilon \to 0$ (a more rigorous algorithm in the case of nondiagonal $\mathbf{C}_0$ would be to remove as much diagonal part of $\mathbf{C}_0$ as possible while keeping the rest still positive semidefinite and apply the small ϵ frequency treatment, we just described, to an eigendecomposition of the nondiagonal part).

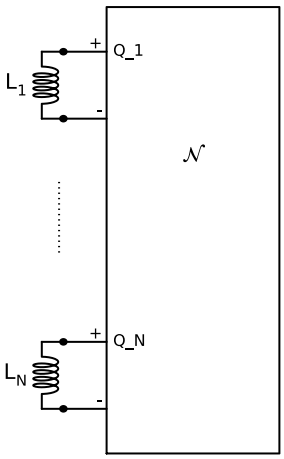


Fig. 2. Multiport blackbox network $\mathcal{N}$ in Fig. 1 with Josephson junctions L_{J_i} replaced with linear inductors L_i. L_i values are related to the bare junction inductances L_{J_i} by $L_i = L_{J_i}/(1 - (2E_C^{(i)}/\hbar\omega_i))$.

$\mathbf{R}$ is an $M \times N$ matrix generating the couplings between qubits and internal modes. $\mathbf{R}$ consists of row vectors $r_k = (r_{k1} \ldots r_{kN})$ with $r_k^T r_k = \mathbf{A}_k$. $\mathbf{M}_0$ matrix is diagonal with entries $(1/L_1, \ldots, 1/L_N, 1/L_{R_1}, \ldots, 1/L_{R_M})$, where $L_{R_k} = 1/\omega_{R_k}^2$ for $1 \le k \le M$. Here, we replaced the Josephson junction i in Fig. 1 with the qubit inductance L_i such that $1/\sqrt{L_i C_i} = \omega_i$, as shown in Fig. 2. An important point to note here is that the choice of L_i over the bare junction inductance L_{J_i} makes the two-body terms (that appear after expanding the nonlinear terms in the junction potentials and normal ordering) in [25, eq. (10)] vanish up to the order of interest here. This is crucial since these terms might contain significant residual couplings between qubit and internal modes. We refer the reader to Appendix E for details.

We do a capacitance rescaling [28] $\mathbf{\Phi}_J \to \mathbf{C}_0^{1/2}\mathbf{\Phi}_J$ to transform the capacitance matrix $\mathbf{C}$ as follows:

$$\mathbf{C} \to \begin{pmatrix} \mathbf{1}_{N\times N} & -\mathbf{C}_0^{1/2}\mathbf{R}^T \\ -\mathbf{R}\mathbf{C}_0^{1/2} & \mathbf{1}_{M\times M} + \mathbf{R}\mathbf{C}_0\mathbf{R}^T \end{pmatrix} \quad (19)$$

and $\mathbf{M}_0$ transforming into the diagonal matrix with entries $(\omega_1^2, \ldots, \omega_N^2, \omega_{R_1}^2, \ldots, \omega_{R_M}^2)$. At this point, we note that the coupling g_{ik} between the qubit mode i and the internal mode k is given by

$$g_{ik} = \frac{\sqrt{\omega_i\omega_{R_k}}}{2} r_{ki}\sqrt{C_i} \quad (20)$$

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where we also note that $r_{ki}\sqrt{C_i}$ is a small parameter, i.e., $r_{ki}\sqrt{C_i} \ll 1$. We then apply the transformation

$$\mathbf{T} = \begin{pmatrix} \mathbf{1}_{N\times N} & \mathbf{C}_0^{1/2}\mathbf{R}^T \\ \mathbf{0}_{M\times N} & \mathbf{1}_{M\times M} \end{pmatrix} \tag{21}$$

to reduce the capacitance matrix to identity

$$\mathbf{C} \to \mathbf{T}^T\mathbf{C}\mathbf{T} = \mathbf{1}. \tag{22}$$

Then, $\mathbf{M}_0$ transforms to $\mathbf{M}_1$ as

$$\mathbf{M}_1 = \mathbf{T}^T\mathbf{C}_0^{-1/2}\mathbf{M}_0\mathbf{C}_0^{-1/2}\mathbf{T} = \begin{pmatrix} \mathbf{\Omega}_J^2 & \mathbf{\Omega}_J^2\mathbf{C}_0^{1/2}\mathbf{R}^T \\ \mathbf{R}\mathbf{C}_0^{1/2}\mathbf{\Omega}_J^2 & \mathbf{\Omega}_{R'}^2 \end{pmatrix} \tag{23}$$

where $\mathbf{\Omega}_{R'}^2 = \mathbf{\Omega}_R^2 + \mathbf{R}\mathbf{C}_0^{1/2}\mathbf{\Omega}_J^2\mathbf{C}_0^{1/2}\mathbf{R}^T$, $\mathbf{\Omega}_J$ is diagonal with entries $(\omega_1, \ldots, \omega_N)$, $\omega_i = 1/\sqrt{L_iC_i}$ for $1 \le i \le N$, and $\mathbf{\Omega}_R$ is diagonal with entries $(\omega_{R_1}, \ldots, \omega_{R_M})$. Here, we observe that the resonator frequencies get small corrections that we will neglect in the following and the couplings in between the modes in the resonator subspace are of order $\Omega_J(g/\Omega_R)^2$, where g is the bare coupling strength between qubit and resonator modes. The resonator subspace being diagonal to order $\Omega_J(g/\Omega_R)^2$ is important in the application of the Schrieffer–Wolff transformation in the following, as it allows to capture small couplings by only a second-order Schrieffer–Wolff transformation that would otherwise require higher order corrections.

We now block diagonalize $\mathbf{M}_1$ by applying a Schrieffer–Wolff transformation to get

$$\widetilde{\mathbf{M}}_1 = \exp(-\mathbf{S})\mathbf{M}_1\exp(\mathbf{S}) \tag{24}$$

where $\mathbf{S}$ is skew symmetric and $\widetilde{\mathbf{M}}_1$ is the block diagonal which can be computed up to desired order in the bare couplings using [29, eqs. (B.12) and (B.15)]. We note that since this transformation is unitary, it will keep the capacitance matrix identity such that we have the following block-diagonal Hamiltonian in the final frame:

$$\mathcal{H} = \frac{1}{2}\boldsymbol{q}^T\boldsymbol{q} + \frac{1}{2}\boldsymbol{\phi}^T\widetilde{\mathbf{M}}_1\boldsymbol{\phi} + \mathcal{O}(\varphi_J^4) \tag{25}$$

where the final coordinate fluxes $\boldsymbol{\phi}$ are related to the initial coordinates $\mathbf{\Phi}$ by

$$\mathbf{\Phi} = \begin{pmatrix} \mathbf{\Phi}_J \\ \mathbf{\Phi}_R \end{pmatrix} = \begin{pmatrix} \mathbf{C}_0^{-1/2} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{pmatrix}\mathbf{T}\exp(\mathbf{S})\boldsymbol{\phi} \tag{26}$$

and $\mathcal{O}(\varphi_J^4)$ term stands for higher order nonlinear corrections giving anharmonicities and dispersive shifts between modes calculated in Appendix E.

Using [29, eq. (B.15c)], to second order in the bare couplings

$$(\widetilde{\mathbf{M}}_1)_{ij} = \frac{1}{2}\sum_k (\mathbf{M}_1)_{ik}(\mathbf{M}_1)_{kj}\left[\frac{1}{\omega_i^2 - \omega_{R_k}^2} + \frac{1}{\omega_j^2 - \omega_{R_k}^2}\right] \tag{27}$$

where again i and j are qubit labels and k labels internal modes. $(\mathbf{M}_1)_{ik}$ is the (i,k)th entry of the matrix $\mathbf{M}_1$, and from (23), we have

$$(\mathbf{M}_1)_{ik} = \omega_i^2 C_i^{1/2} r_{ki}. \tag{28}$$

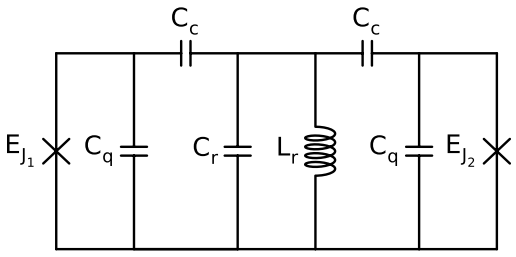


Fig. 3. Example circuit of two transmons capacitively coupled through a single-mode bus. Both transmons have the same shunting capacitance C_q and the same coupling capacitances C_c to the bus.

Noting again $r_k^T r_k = \mathbf{A}_k$, we can write

$$\begin{aligned}(\mathbf{M}_1)_{ik}(\mathbf{M}_1)_{kj} &= \omega_i^2\omega_j^2 C_i^{1/2}C_j^{1/2} r_{ki} r_{kj} \\ &= \omega_i^2\omega_j^2 C_i^{1/2}C_j^{1/2}(\mathbf{A}_k)_{ij}.\end{aligned} \tag{29}$$

Hence, we can rewrite (27) as

$$\begin{aligned}(\widetilde{\mathbf{M}}_1)_{ij} &= \frac{1}{2}\sum_k \omega_i^2\omega_j^2 C_i^{1/2}C_j^{1/2}\left[\frac{(\mathbf{A}_k)_{ij}}{\omega_i^2 - \omega_{R_k}^2} + \frac{(\mathbf{A}_k)_{ij}}{\omega_j^2 - \omega_{R_k}^2}\right] \\ &= -\frac{1}{2}\omega_i^2\omega_j^2 C_i^{1/2}C_j^{1/2}\mathrm{Im}\left[\frac{Z_{ij}(\omega_i)}{\omega_i} + \frac{Z_{ij}(\omega_j)}{\omega_j}\right].\end{aligned} \tag{30}$$

Quantizing the system by introducing the annihilation and creation operators for the qubit modes in the final frame by $\hat{\phi}_i = (\hbar Z_i/2)^{1/2}(\hat{b}_i + \hat{b}_i^\dagger)$ for $1 \le i \le N$ and noting that the characteristic impedance Z_i for the qubit mode i is $Z_i = 1/\omega_i$ in that frame

$$\begin{aligned}J_{ij} &= \frac{1}{2}\sqrt{Z_iZ_j}(\widetilde{\mathbf{M}}_1)_{ij} \\ &= -\frac{1}{4}\sqrt{\frac{\omega_i\omega_j}{L_iL_j}}\mathrm{Im}\left[\frac{Z_{ij}(\omega_i)}{\omega_i} + \frac{Z_{ij}(\omega_j)}{\omega_j}\right].\end{aligned} \tag{31}$$

J_{ij} in (31) is in the units of radians per second. We note that this formula takes into account all the modes (possibly infinite) that might be present in the electromagnetic structure coupling the qubits.

A. Example 1: Two Transmons Coupled Through a Single-Mode LC Resonator Bus

In this section, we will apply the formula in (5) for the J-coupling rate derived in the previous section to the simple circuit of two transmons coupled through a lumped LC resonator, as shown in Fig. 3, and compare it to the expression derived in [30]

$$J = \frac{g_1g_2(\omega_1 + \omega_2 - 2\omega_r)}{2(\omega_1 - \omega_r)(\omega_2 - \omega_r)} \tag{32}$$

where g_1 and g_2 are the couplings of qubits 1 and 2 to the bus, and ω_1, ω_2, and ω_r are the qubit and resonator frequencies, respectively.

The circuit in Fig. 3 has the following Hamiltonian:

$$H = \frac{1}{2}\mathbf{Q}^T\mathbf{C}^{-1}\mathbf{Q} + \frac{1}{2}\mathbf{\Phi}^T\mathbf{M}_0\mathbf{\Phi} - E_{J_1}\cos(\varphi_1) - E_{J_2}\cos(\varphi_2) \tag{33}$$

where

$$\mathbf{C} = \begin{pmatrix} C_q + C_c & 0 & -C_c \\ 0 & C_q + C_c & -C_c \\ -C_c & -C_c & C_r + 2C_c \end{pmatrix}. \tag{34}$$

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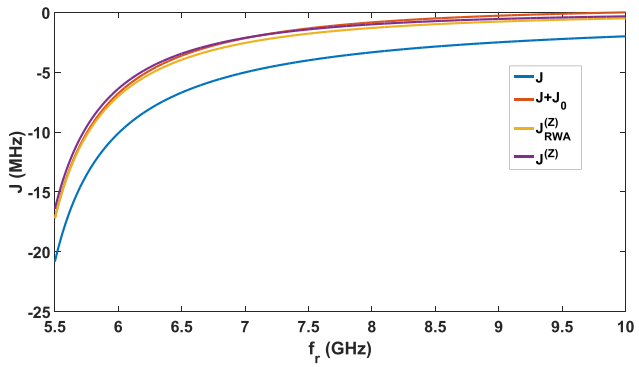


Fig. 4. Comparison of J-coupling expressions $J^{(Z)}$, $J^{(Z)}_{\mathrm{RWA}}$, J, and $J+J_0$ for bus frequency f_r ranging from 5.5 to 10 GHz for the circuit in Fig. 3 with the following set of parameter values $g_1 = g_2 = 100$ MHz, $\omega_1 = 2\pi$ (4.90 GHz) and $\omega_2 = 2\pi$ (5.10 GHz), and $\delta_1 = \delta_2 = -340$ MHz. Vertical axis is the J-coupling rate in megahertz.

$\mathbf{M}_0$ diagonal with entries $(0, 0, 1/L_r)$ and the coordinate vector $\mathbf{\Phi} = (\Phi_{J_1}, \Phi_{J_2}, \Phi_r)^T$ holds the fluxes across the inductive branches. Typically, $C_c \ll C_q \ll C_r$ holds so that we can approximately write

$$\mathbf{C}^{-1} \cong \begin{pmatrix} 1/C_q & \dfrac{C_c^2}{C_q^2 C_r} & \dfrac{C_c}{C_q C_r} \\ \dfrac{C_c^2}{C_q^2 C_r} & 1/C_q & \dfrac{C_c}{C_q C_r} \\ \dfrac{C_c}{C_q C_r} & \dfrac{C_c}{C_q C_r} & 1/C_r \end{pmatrix} \tag{35}$$

so that we have

$$g_i = \frac{1}{2\sqrt{Z_i Z_r}} \frac{C_c}{C_q C_r} \tag{36}$$

where $Z_i = \sqrt{L_i/C_q}$ and $Z_r = \sqrt{L_r/C_r}$. We note here that although there is no direct electrostatic dipole coupling between qubits in (34), a mediated coupling J_0 appears in (35). As we will see in the following, the magnitude of J_0 is nonnegligible compared to J in (32); hence, one should compute $J + J_0$ for the total exchange coupling rate, as we did in Fig. 4. We note that

$$J_0 = \frac{1}{2\sqrt{Z_1 Z_2}} \frac{C_c^2}{C_q^2 C_r} = \frac{2}{\omega_r} g_1 g_2. \tag{37}$$

We now apply the impedance formula for the J-coupling in (5) to the circuit in Fig. 3. We need to first compute the two-port impedance matrix between the ports shunted by Josephson junctions. This can be done by an $ABCD$-matrix analysis [31], for example. One then gets

$$\mathrm{Im}[Z_{12}(\omega)] = \frac{C_c^2 L_r \omega/(C_q + C_c)}{C_q(1 - \omega^2/\omega_r^2) + C_c\left(1 - 2\omega^2/\omega_{qr}^2 - \omega^2/\omega_r^2\right)} \tag{38}$$

where $\omega_r = 1/\sqrt{L_r C_r}$ and $\omega_{qr} = 1/\sqrt{L_r C_q}$. We note that in actual devices, $C_q \ll C_r$, and hence, $\omega_r \ll \omega_{qr}$. We can then neglect the term $-2\omega^2/\omega_{qr}^2$ appearing in the denominator compared to the term $-\omega^2/\omega_r^2$ such that

$$\mathrm{Im}[Z_{12}(\omega)] \cong \frac{C_c^2 L_r \omega}{(C_q + C_c)^2\left(1 - \omega^2/\omega_r^2\right)}. \tag{39}$$

TABLE I

MEASURED QUBIT FREQUENCIES f_q AND ANHARMONICITIES δ IN THE IBM Q 16 RUESCHLIKON DEVICE SHOWN IN FIG. 5

Q	1	2	3	4	5	6	7	8
$f_q\,(GHz)$	5.3961	5.2756	5.0831	4.9791	5.1513	5.3058	5.2528	5.1153
$\delta\,(MHz)$	-291.0	-289.2	-294.0	-290.7	-286.0	-289.2	-298.4	-287.6
Q	0 (16)	15	14	13	12	11	10	9
$f_q\,(GHz)$	5.2561	5.1095	4.8701	5.0881	4.9466	5.1107	5.0426	5.1555
$\delta\,(MHz)$	-292.8	-289.8	-296.0	-289.8	-299.0	-290.7	-291.5	-297.6

Noting also $C_c \ll C_q$, we have

$$\begin{aligned} \mathrm{Im}[Z_{12}(\omega)] &\cong \frac{C_c^2 L_r \omega}{C_q^2\left(1 - \omega^2/\omega_r^2\right)} \\ &= \frac{1}{2} \frac{C_c^2 L_r \omega_r \omega}{C_q^2} \left(\frac{1}{\omega_r - \omega} + \frac{1}{\omega_r + \omega}\right) \end{aligned} \tag{40}$$

hence, by (5)

$$\begin{aligned} J^{(Z)} = -\frac{1}{8} \frac{\sqrt{\omega_1 \omega_2}}{\sqrt{L_1 L_2}} \frac{C_c^2 L_r \omega_r}{C_q^2} \Bigg(&\frac{1}{\omega_r - \omega_1} + \frac{1}{\omega_r - \omega_2} \\ &+ \frac{1}{\omega_r + \omega_1} + \frac{1}{\omega_r + \omega_2}\Bigg) \end{aligned} \tag{41}$$

where we used the superscript Z to indicate the application of the impedance J-coupling formula in (5).

If we interpret the first two terms inside the parenthesis in (41) as the RWA-terms, we can write

$$\begin{aligned} J^{(Z)}_{\mathrm{RWA}} &= -\frac{1}{8} \frac{\sqrt{\omega_1 \omega_2}}{\sqrt{L_1 L_2}} \frac{C_c^2 L_r \omega_r}{C_q^2} \left(\frac{1}{\omega_r - \omega_1} + \frac{1}{\omega_r - \omega_2}\right) \\ &= \left(\frac{\omega_1 \omega_2}{\omega_r^2}\right) \frac{g_1 g_2 (\omega_1 + \omega_2 - 2\omega_r)}{2(\omega_1 - \omega_r)(\omega_2 - \omega_r)} \\ &= \left(\frac{\omega_1 \omega_2}{\omega_r^2}\right) J. \end{aligned} \tag{42}$$

We note here that the standard expression for the exchange coupling J in (32) is obtained with an RWA; this is why we only kept the first two terms inside the parenthesis in (41) and defined $J^{(Z)}_{\mathrm{RWA}}$ in (42).

We now compare the formulas obtained in Fig. 4 with the following set of realistic parameter values $g_1 = g_2 = 100$ MHz, $\omega_1 = 2\pi$ (4.90 GHz) and $\omega_2 = 2\pi$ (5.10 GHz), and $\delta_1 = \delta_2 = -340$ MHz.

III. EXPERIMENTAL VALIDATION OF THE EXCHANGE COUPLING FORMULA

To validate our result in (5), we compare measurements taken in our labs from the multiqubit device shown in Fig. 5 with the full-wave FEM electromagnetic simulations performed using Ansys HFSS. This multiqubit device is freely accessible through the IBM Quantum Experience cloud computing platform. The chip contains 16 qubits labeled by Q1–Q16(0) which are arranged in the 2×8 configuration with a total of 22 coplanar waveguide (CPW) half-wave resonator busses connecting the qubits as shown in Fig. 6. Each bus connects two qubits. R1–R16(0) in Fig. 5 label the readout resonators, which are also CPW half-wave resonators. Measured qubit frequencies and anharmonicities are listed in Table I.

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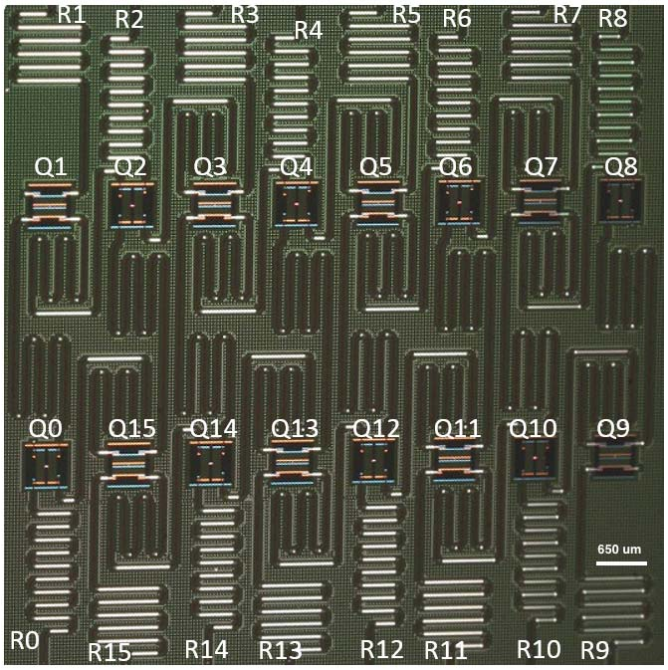


Fig. 5. IBM Q 16 Rueschlikon Device. This is a 16 qubit chip with qubits arranged in the 2×8 configuration with bus connectivity, as shown in Fig. 6. There are a total of 22 CPW half-wave resonator buses with two qubits per bus. Qubits are labeled by Q1–Q16(0). Readout resonators are labeled by R1–R16(0). Readout resonators are also CPW half-wave resonators with frequencies around 7 GHz.

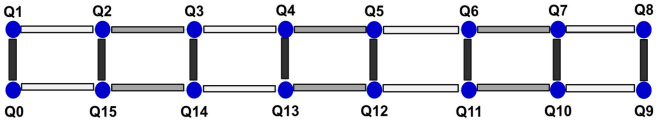


Fig. 6. Connectivity of the IBM Q 16 Rueschlikon Device; 16 qubits are arranged in two rows in the 2×8 configuration with 22 CPW half-wave bus resonators connecting the qubits. There are two qubits per bus and three distinct bus frequencies: 6.25 GHz (white bars), 6.45 GHz (gray bars), and 6.65 GHz (black bars)

We back up exchange coupling rates J_M between qubits connected by a bus in the device in Fig. 5 from the experimental ZZ-rate measurements, as detailed in Appendix C. We also calculate exchange coupling rates J_S between the pairs of qubits by applying (5) using experimentally measured qubit frequencies and anharmonicities listed in Table I together with FEM electromagnetic simulations of the off-diagonal impedance entries Z_{ij}, as detailed in Appendix C. The scatter plot of J_M and J_S values around the $y = x$ line is shown in Fig. 7. We note that the largest deviations from the $y = x$ line occur for the qubit pairs that are on the edge of the device in Figs. 5 and 6. This might be due to spurious couplings of qubits that are on the edge to some boundary modes developed around the chip that is not modeled in our simulations. Note also that the edge qubits are only connected to two bus resonators.

A. Example 2: Scaling of J Coupling Rates in the Multiqubit device

In this section, we apply the impedance formula in (5) for the exchange couplings J_{ij} to the multiqubit device shown

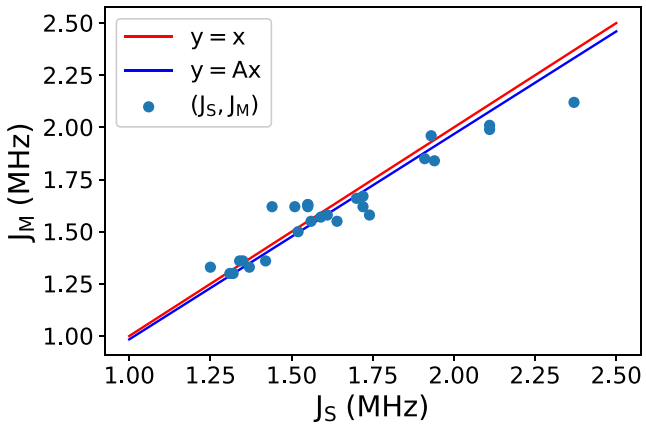


Fig. 7. Scatter plot of the exchange coupling values around the $y = x$ line (red line). J_M are the values obtained from the ZZ-measurements on the multiqubit device shown in Fig. 5 and J_S are the values obtained by the application of (5) as explained in detail in Appendix C. We observe good agreement between the experimental measurements and the theory. The largest deviation occurs from the $y = x$ line for the qubit pairs that are on the edge of the chip shown in Figs. 5 and 6. Blue line is the least squares fit to $y = Ax$ with $A = 0.984$ and RMSE $= 0.092$.

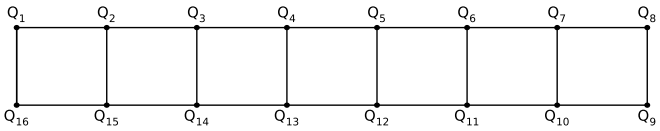


Fig. 8. 2×8 device connectivity: 16 qubits are arranged in two rows. Nodes represent the qubits, while edges linking the qubits represent busses. There are two qubits connected to each bus and there is a total of 22 busses.

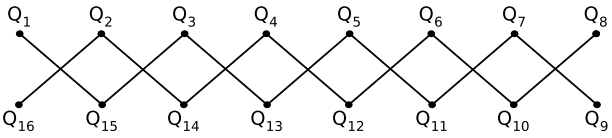


Fig. 9. 2×8 device with four-qubit-per-bus arrangement. Scaling of the J couplings over the lattice is compared in Fig. 10 to the arrangement in Fig. 8. Crossed links represent bus resonators each connected to four qubits and there are seven buses in total.

in Fig. 8 to calculate the decay of the J-coupling over the chip. This is a simplified lumped element circuit model of the multiqubit device shown in Fig. 5 recently released by IBM in its online cloud environment for quantum computing: IBM Q Experience. The device consists of 16 qubits arranged in two rows and connected to each other by 22 bus resonators with two qubits per bus. To compare, we also apply the impedance formula for J-coupling to the arrangement shown in Fig. 9 where we have four qubits on each bus.

Fig. 10 is generated from a lumped element circuit model of the 2×8 multiqubit device. The bus connections in Fig. 8 are as given in Fig. 3 for the two-qubit-per-bus arrangement with the qubits capacitively coupled to the LC-resonator bus. In the four-qubit-per-bus arrangement shown in Fig. 9, four qubits are capacitively coupled to a single LC-resonator in a similar way. In both the cases, all qubits are assumed to be at 5.10 GHz, and the coupling capacitor values C_c in Fig. 3 are 7.5 fF. The bus frequencies are all equal to 6.30 GHz. These parameter values do not correspond directly to any real physical device that has been measured in our labs. They have been rather chosen as approximate parameter values for the

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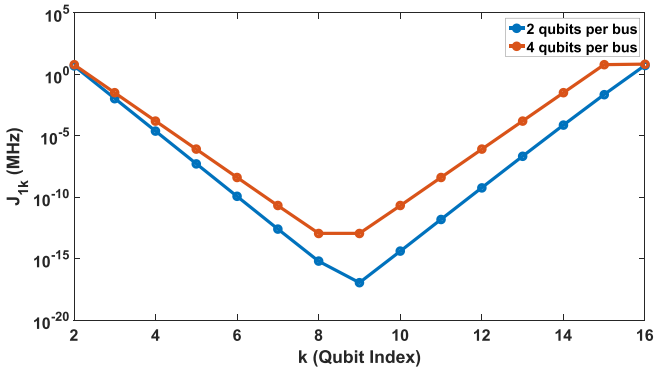


Fig. 10. J_{1k} coupling rate for $k = 2, \ldots, 16$ as measured from the first qubit Q_1 in Figs. 8 and 9 as a function of qubit index k. The exchange coupling rate J_{1k} decays exponentially for $k = 2, \ldots, 8$. After k = 9, J_{1k} shows an increase for both types of chips since we are getting closer to the first qubit traversing the bottom row. J_{1k} is the same for $k = 8$ and $k = 9$ in the four-qubit-per-bus arrangement since the shortest path from qubit 1 is the same. Similarly, J_{1k} is the same for $k = 15$ and $k = 16$ in the four-qubit-per-bus arrangement since qubits 15 and 16 are coupled through the same bus to the qubit 1. J_{12} is -4.9 and -6.1 MHz in the devices in Figs. 8 and 9, respectively.

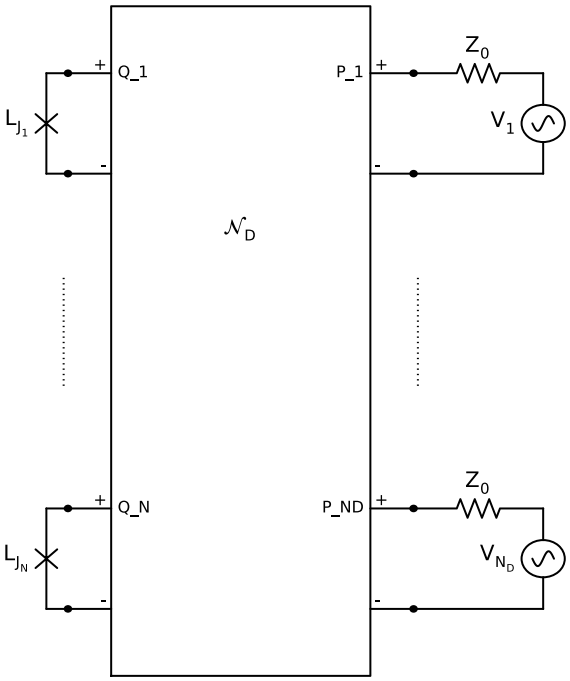


Fig. 11. Multiport circuit in Fig. 1 augmented with drives. Drive ports are labeled by P_1 to P_ND.

numerical study of the scaling of the J-coupling over the 2×8 multiqubit device.

IV. Couplings of the Qubits to the Voltage Drives

Qubits are coupled to room temperature electronics for their readout and control. Readout and control signals pass through several amplification/attenuation stages, as they travel through different stages in a dilution fridge. In between these stages, they are carried over transmission lines, such as the coaxial cables or the striplines, on a printed circuit board (PCB). We will content ourselves here with modeling this coupling mechanism simply by voltage sources driving the quantum chip through transmission lines (which we assume to be infinite in extent to keep things simple here and represent them simply by resistors Z_0), as shown in Fig. 11. This circuit is an augmented version of the multiport canonical circuit in Fig. 1, where N_D "drive ports" are added. The drive ports are defined at positions where drive lines reach the chip (see Appendix D for more details on how to define the drive ports in a 3-D finite-element electromagnetic simulator). They are connected to transmission lines of characteristic impedance Z_0 (typically $Z_0 = 50\ \Omega$), which, in turn, are shunted by the voltage sources V_d for $1 \leq d \leq N_D$. Such a simple circuit model will allow us to derive expressions for the couplings ε_{id} of the qubits to voltage drives in this section. A similar analysis in Section V will allow us to compute Purcell loss rates of the qubit modes due to their coupling to the drive lines.

As we will see in Appendix B, the system has the following Hamiltonian given in (101) in the final block-diagonalized frame corresponding to $\widetilde{\mathbf{M}}_1$ in (25):

$$H = \frac{1}{2}(\boldsymbol{q} - \mathbf{C}_q * \mathbf{V}_V)^T(\boldsymbol{q} - \mathbf{C}_q * \mathbf{V}_V) + \frac{1}{2}\boldsymbol{\phi}^T \widetilde{\mathbf{M}}_1 \boldsymbol{\phi} + \mathcal{O}(\boldsymbol{\varphi}_J^4) \tag{43}$$

where the $(N+M) \times N_D$ matrix $\mathbf{C}_q$ gives the coupling of the voltage sources $\mathbf{V}_V = (V_1, \ldots, V_{N_D})$ to the charge degrees of freedom $\boldsymbol{q}$ of the circuit and $*$ is the time-convolution operator. After quantizing this Hamiltonian by introducing the harmonic mode operators $\hat{q}_i = -i(\hbar/2Z_i)^{1/2}(\hat{b}_i - \hat{b}_i^\dagger)$ for the qubit modes and computing the projection of $\mathbf{C}_q$ onto the qubit subspace, one obtains the following drive term acting in the qubit subspace:

$$H_{id}^D = i\sqrt{\frac{\hbar\omega_i}{2L_i}}\mathrm{Im}[Z_{i,p(d)}(\omega_i)]\frac{C_{p(d)}V_d(\hat{b}_i - \hat{b}_i^\dagger)}{1 + i\omega_d Z_0 C_{p(d)}} \tag{44}$$

from which we get

$$\varepsilon_{id} = \sqrt{\frac{\omega_i}{2\hbar L_i}}\mathrm{Im}[Z_{i,p(d)}(\omega_i)]\frac{e^{i\theta_d}C_{p(d)}}{\sqrt{1 + \omega_d^2 Z_0^2 C_{p(d)}^2}} \tag{45}$$

for the coupling matrix ε_{id} appearing in (6) and giving the coupling of the qubit modes to the voltage drives. Here, $\theta_d = (\pi/2) - \arctan(\omega_d Z_0 C_{p(d)})$ and $Z_{i,p(d)}$ is the impedance entry connecting the qubit port i to drive port [with port index $p(d)$] corresponding to the voltage source V_d. $C_{p(d)}$ is the total capacitance shunting the dth drive port, ω_d is the frequency of the signal driving the qubit j, and Z_0 is the characteristic impedance of the drive lines (typically $Z_0 = 50\Omega$). The last factor in (44) is just a voltage division factor giving how much of the drive voltage V_d is seen across the dth drive port. The factor $\mathrm{Im}[Z_{i,p(d)}(\omega_i)]$ gives, on the other hand, the classical crosstalk.

A. Classical Crosstalk and the Location of the Drive Ports

We define the classical crosstalk as the unwanted drive the qubit i experiences when we excite the device only through the drive line of the qubit j. For the purpose of understanding the classical crosstalk, we will be only interested in the relative magnitudes of the voltages seen by different qubits and according to the analysis in Appendix B

$$X_{ij} = 20\log_{10}\left(\left|\frac{\mathrm{Im}[Z_{i,d(j)}(\omega_i)]}{\mathrm{Im}[Z_{j,d(j)}(\omega_j)]}\right|\right) \tag{46}$$

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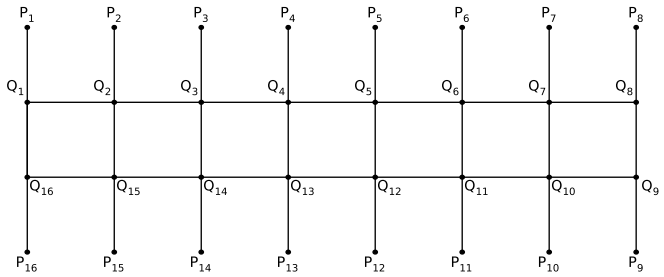


Fig. 12. 2×8 device model with two-qubit-per-bus connectivity in Fig. 8 with readout resonators and drive ports added. Readout resonators are represented with edges linking qubits Q_k to their drive ports P_k.

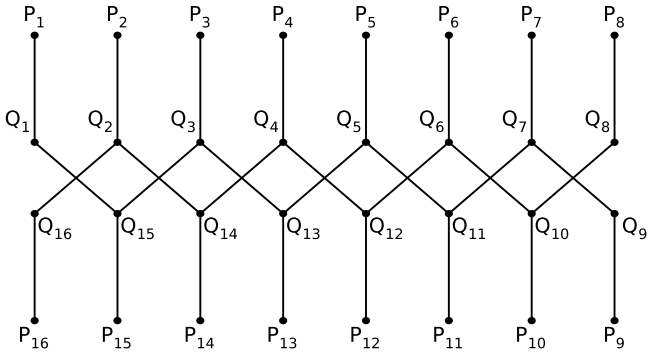


Fig. 13. 2×8 device model with four-qubit-per-bus connectivity in Fig. 9 with readout resonators and drive ports added. Readout resonators are represented with edges linking qubits Q_k to their drive ports P_k.

is a good measure of the classical crosstalk in the units of dB. Here, $Z_{i,d(j)}(\omega_i)$ is the impedance entry connecting the drive port $d(j)$ of the qubit j to the qubit port i.

Although we have already stated in the previous sections that we defined the drive ports where the drive lines reach the chip, we give a more precise description here on how we choose the locations of the drive ports. As the drive signals travel over the transmission lines toward the chip, they will eventually reach the transition region (before launching onto the chip) where they will no longer see a constant impedance but a discontinuity of which some portion of the signal will be reflected back. Ideally, one would like to define the drive ports at positions where this discontinuity first starts to appear. The exact positions can be determined with a time-domain reflectometry (TDR) measurement/simulation for example. In the absence of TDR information, one can make a safe choice by keeping the drive ports far enough from the chip boundary. In electromagnetic finite-element simulators, such ports will typically be defined as wave ports on the planes (perpendicular to the direction of propagation) in the cross sections of the drive lines. Such a choice for the drive ports will include any crosstalk happening in the transition region (such as a spurious chip boundary mode [32] for example) in the crosstalk measure defined in (46). See Appendix D for more details on how to define the drive ports in electromagnetic finite-element simulators.

B. Example: Classical Crosstalk in the Multiqubit Device

In this section, we augment our models for the 2 × 8 multiqubit devices by adding the readout resonators and the drive ports, as shown in Figs. 12 and 13, and apply the the formula in (46) to evaluate the crosstalk in the device. We plot X_{1k} which gives the crosstalk between the drive line of the qubit Q_1 and the other qubits in Figs. 12 and 13 as a function of the qubit label $k = 2, \ldots, 16$ in Fig. 14. The plot in Fig. 14 is generated using the lumped element model for the coupling of the qubits to the readout resonators and the coupling of the readout resonators to the 50-Ω lines, as shown in Fig. 15. The readout resonators are assumed to be at 7 GHz and the coupling capacitors in Fig. 15 are $C_c = 3$ fF, $C_\kappa = 2.5$ fF, and $C_d = 100$ fF. Bus connections are as described in Section III-A.

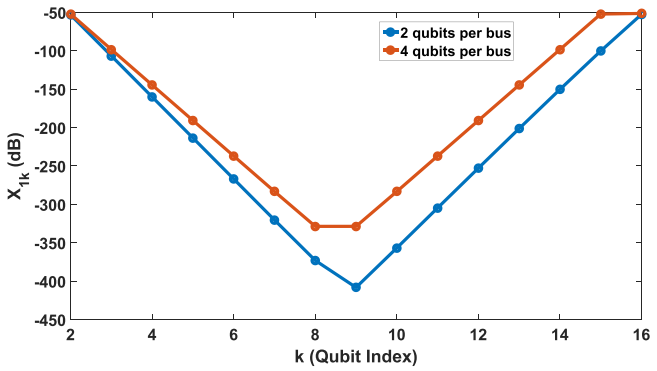


Fig. 14. How much of the drive voltage leaks into other qubits in the 2 × 8 multiqubit devices in Figs. 12 and 13 when only the qubit Q_1 is excited through its drive port P_1; classical crosstalk decays exponentially as a function of distance from the qubit Q_1 to the right in the first row of the circuit up to $k = 8$. The pattern agrees with the the J-coupling pattern in Fig. 10.

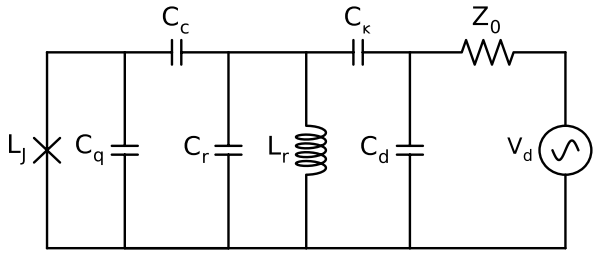


Fig. 15. Example circuit of a Transmon qubit coupled to a readout resonator which in turn coupled to external drive line of characteristic impedance Z_0. C_d is the shunt capacitance of the drive port.

V. Purcell Loss Rates of the Qubit Modes

Qubits are coupled to external electronics for their readout and control. In Section IV, we analyzed the couplings of the qubits to voltage drives. The same coupling mechanism causes relaxation of the excitations in the qubit modes which is called the "Purcell Loss." In this section, we compute rates for the Purcell loss of the qubit modes we identified in the earlier sections.

As in Section IV, the coupling of the qubits to external electronics is modeled with the idealized circuit model in Appendix B, and we will use the same coupling matrices of the formalism in [19] that we calculated in Appendix B for the drive couplings. We have N_D baths corresponding to transmission lines of characteristic impedances Z_0 driving the qubits. Assuming couplings of qubits to the lines are small, to first order in these couplings, we will assume that T_1 rates can be computed separately for each bath. The total rate will then be the sum of rates due to each line.

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We start by noting that when we have only the bath due to the drive line of the voltage source V_d with port index $p(d)$, $\mathbf{C}_D$ defined in (88) is a scalar $C_{p(d)}$ for $1 \le d \le N_D$. Hence, $\bar{\mathbf{m}}$ in (92) is

$$\bar{\mathbf{m}}_d = -C_{p(d)} \begin{pmatrix} \mathbf{0}_{N\times 1} \\ \mathbf{v}_d \end{pmatrix} \tag{47}$$

where $\mathbf{v}_d = (\, v_{1d} \ \ldots \ v_{Md} \,)^T$ is the dth column of the matrix $\mathbf{V}$ corresponding to the drive line with port index $p(d)$. After the Schrieffer–Wolff transformation by (99)

$$m_{id} = C_{p(d)} \mathrm{Im}[Z_{i,p(d)}(\omega_i)]/\sqrt{L_i} \tag{48}$$

where m_{id} is the coupling of the bath due to the dth drive line to the qubit mode i.

We now need to compute the spectral densities of the baths corresponding to the transmission lines. $\bar{\mathbf{C}}_Z(\omega)$ matrix defined in (87) is also a scalar in the case of a single bath corresponding to the dth drive line and is given by

$$\bar{C}_{Z,d}(\omega) = -\frac{i\omega Z_0}{1 + i\omega C_{p(d)} Z_0}. \tag{49}$$

Kernel of the bath due to the dth drive line is given in [19, eq. (35)] as

$$K_d(\omega) = \frac{\bar{C}_{Z,d}(\omega)}{1 + \bar{\mathbf{m}}_d^T \mathbf{C}^{-1} \bar{\mathbf{m}}_d \bar{C}_{Z,d}(\omega)}. \tag{50}$$

The term $\bar{\mathbf{m}}_d^T \mathbf{C}^{-1} \bar{\mathbf{m}}_d$ can be evaluated in the final frame using (48) and noting that $\mathbf{C} = \mathbf{1}$ in the final frame. Hence

$$\bar{\mathbf{m}}_d^T \mathbf{C}^{-1} \bar{\mathbf{m}}_d = C_{p(d)}^2 \sum_i (\mathrm{Im}[Z_{i,p(d)}(\omega_i)])^2 / L_i. \tag{51}$$

The spectrum of the bath is given by

$$\begin{aligned} J_d(\omega) &= -\mathrm{Im}[K_d(\omega)] \\ &= \frac{\omega Z_0}{1 + \omega^2 Z_0^2 \left(C_{p(d)} + \bar{\mathbf{m}}_d^T \mathbf{C}^{-1} \bar{\mathbf{m}}_d\right)^2} \\ &\simeq \frac{\omega Z_0}{1 + \omega^2 Z_0^2 C_{p(d)}^2} \end{aligned} \tag{52}$$

assuming $(\bar{\mathbf{m}}_j^T \mathbf{C}^{-1} \bar{\mathbf{m}}_j)/C_{p(d)} \ll 1$ which holds for typical parameter values and frequencies.

Finally, T_1 rate of the qubit mode i due to the dth drive line can be calculated using [19, eq. (44)]

$$\frac{1}{T_1^{i,d}} = \frac{4}{\hbar} |\langle 0 | m_{id} \hat{\mathbf{Q}}_i | 1_i \rangle|^2 J_d(\omega_i) \coth\left(\frac{\hbar\omega_i}{2k_B T}\right) \tag{53}$$

which can be simplified assuming $\coth(\hbar\omega_i/2k_BT) \simeq 1$ for the typical chip temperatures as

$$\begin{aligned} \frac{1}{T_1^{i,d}} &= \frac{4}{\hbar} |\langle 0 | m_{id} \hat{\mathbf{Q}}_i | 1_i \rangle|^2 J_d(\omega_i) \\ &= \frac{2}{L_i} \mathrm{Im}[Z_{i,p(d)}(\omega_i)]^2 \frac{\omega_i^2 Z_0 C_{p(d)}^2}{1 + \omega_i^2 Z_0^2 C_{p(d)}^2}. \end{aligned} \tag{54}$$

To see that Purcell rates $(1/T_1^{i,d})$ are independent of the drive port shunt capacitances $C_{p(d)}$, we workout $Z_{i,p(d)}(\omega_i)$ for the example circuit shown in Fig. 15. Assuming $C_r \gg C_\kappa$, $C_q \gg C_c$, and $C_d \gg C_\kappa$, one can show that

$$\mathrm{Im}[Z_{12}(\omega)] \simeq \frac{\left(\frac{C_c C_\kappa}{C_q C_d}\right) L_r \omega}{1 - (\omega/\omega_r)^2} \tag{55}$$

with $\omega_r = 1/\sqrt{L_r C_r}$ and port 1 being defined across the Josephson junction and port 2 across C_d so that the Purcell rate $(1/T_1^{i,j})$ derived in (54) gives

$$\begin{aligned} 1/T_1 &= \frac{2}{L_q} \mathrm{Im}[Z_{12}(\omega_q)]^2 \frac{\omega_q^2 Z_0 C_d^2}{1 + \omega_q^2 Z_0^2 C_d^2} \\ &= \frac{2}{L_q} \left(\frac{C_c C_\kappa}{C_q C_r}\right)^2 \frac{(\omega_q/\omega_r)^4}{[1 - (\omega_q/\omega_r)^2]^2} \left(\frac{Z_0}{1 + \omega_q^2 Z_0^2 C_d^2}\right) \end{aligned} \tag{56}$$

where ω_q is the qubit frequency and L_q is the qubit inductance. The expression in (56) will be independent of C_d, the total shunt capacitance of the drive port, in the limit of $\omega_d^2 Z_0^2 C_d^2 \ll 1$, which holds for typical parameter values in the actual experiments.

One can similarly calculate coupling of the qubit to the voltage source V_d using (45) and (55) to get

$$\begin{aligned} \varepsilon_{12} &= e^{i\theta_2} \sqrt{\frac{\omega_q}{2\hbar L_q}} \mathrm{Im}[Z_{12}(\omega_q)] C_d \\ &= \frac{1}{\sqrt{2\hbar Z_q}} \left(\frac{C_c C_\kappa}{C_q C_r}\right) \frac{(\omega_q/\omega_r)^2}{1 - (\omega_q/\omega_r)^2} \left(\frac{e^{i\theta_2}}{\sqrt{1 + \omega_d^2 Z_0^2 C_d^2}}\right) \end{aligned} \tag{57}$$

where $\theta_2 = (\pi/2) - \arctan(\omega_d Z_0 C_d)$ and $Z_q = \sqrt{L_q/C_q}$. Equation (57) for the coupling ε_{12} of the qubit to its voltage source V_d will be again independent of C_d in the limit of $\omega_d^2 Z_0^2 C_d^2 \ll 1$, which holds for typical parameter values.

VI. Expressions for the Qubit Anharmonicities and the Dispersive Shifts in the Resonator Frequencies

In this section, we derive expressions for the anharmonicity δ_i of the qubit mode i and dispersive shift χ_{ik} in the frequency ω_{R_k} of the resonator mode k due to qubit mode i using the results of Appendix E. Anharmonicities and dispersive shifts are generated by the nonlinear terms in the expansion of the junction potentials.

From the term H_β in (115) in the expansion in (113) originally given in [25], we note the following:

$$\delta_i = -12\beta_{iiii} \tag{58}$$

$$\chi_{ik} = -24\beta_{iikk}. \tag{59}$$

Using the expression for $\beta_{pp'qq'}$ in (126) and (120) and (122), we obtain

$$\delta_i = -E_C^{(i)} \left(\frac{\omega_{J_i}}{\omega_i}\right)^2 \tag{60}$$

$$\chi_{ik} = -2E_C^{(i)} \left(\frac{\omega_{J_i}^2}{\omega_i \omega_{R_k}}\right) r_{ki}^2 C_i \left(\frac{\omega_{R_k}^2}{\omega_{R_k}^2 - \omega_i^2}\right)^2. \tag{61}$$

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From (18), we note

$$r_{ik} = \frac{2g_{ik}}{\sqrt{C_i \omega_i \omega_{R_k}}}. \tag{62}$$

Hence

$$\chi_{ik} = -2E_C^{(i)} \left(\frac{\omega_{J_i}^2}{\omega_i \omega_{R_k}} \right) \left(\frac{4g_{ik}^2}{\omega_i \omega_{R_k}} \right) \left(\frac{\omega_{R_k}^2}{\omega_{R_k}^2 - \omega_i^2} \right)^2 = 8\delta_i \left(\frac{g_{ik}\omega_{R_k}}{\omega_{R_k}^2 - \omega_i^2} \right)^2. \tag{63}$$

Since dispersive shifts χ_{ik} are usually small, one might need to include the sixth-order terms in the expansion of junction potentials for a more accurate formula than (63). This can be done using the expansions in (114) and (115) in a recursive way with simple counting arguments to get the following formula with a correction factor:

$$\chi_{ik} = 8\delta_i \left(1 - \frac{6E_C^{(i)}}{\omega_i} \right) \left(\frac{g_{ik}\omega_{R_k}}{\omega_{R_k}^2 - \omega_i^2} \right)^2. \tag{64}$$

VII. Conclusion

We have analyzed superconducting quantum processors consisting of low anharmonicity transmon qubits. We have shown that the exchange coupling rates between qubits are related in a simple way to the off-diagonal entry of the multiport impedance matrix connecting the qubit ports evaluated at qubit frequencies. Qubit ports are defined across the Josephson junctions. Similarly, coupling of the qubits to their drives and Purcell relaxation rates of the qubit modes are related to the entry of the multiport impedance matrix connecting the qubits and the drive ports. This gives a complete microwave description of the system in the qubit subspace. The formulas requiring only evaluation at qubit frequencies (no need for frequency sweeps and fitting) make modeling and simulation of the chips much more efficient.

Simple relations of the qubit exchange coupling rates and the couplings of the qubits to the voltage drives to the impedance response allow application of microwave engineering techniques to improve the performance of the two-qubit gates. One application could be to use the microwave coupler or filtering structures to shape the response profile to reduce unwanted terms in two-qubit gates.

Appendix A
Derivation of the Hamiltonian for the Canonical Multiport Cauer Circuit

Any multiport lossless impedance response can be synthesized with the canonical Cauer circuit shown in Fig. 16. The Cauer circuit consists of N "qubit ports" on the left shunted by the Josephson junctions in our case and M internal modes synthesized as parallel LC tank circuits on the right. Couplings between the ports and the internal modes are mediated by the multiport Belevitch transformers (see [22] for details). A purely capacitive stage [see Fig. 16 (top right)] provides total shunt capacitances of the junctions. In the most general form shown in Fig. 16, there is a purely inductive stage shown in the bottom-right corner. This stage is responsible for the purely inductive energy storage in the system. However, in most of the physical situations arising with distributed electromagnetic structures, this stage will be absent since any distributed inductor will always have a finite parasitic capacitance. For cases where such a stage is really necessary, the degrees of freedoms associated with it can be eliminated with a Born–Oppenheimer analysis [28].

The synthesis of the canonical Cauer circuit in Fig. 16 proceeds as follows: first, we do the eigendecomposition of the residue at dc $\mathbf{A}_0$ in (16):

$$\mathbf{A}_0 = \mathbf{U}\mathbf{C}_0^{-1}\mathbf{U}^T \tag{65}$$

where $\mathbf{U}$ is the $N \times N$ orthonormal matrix holding the eigenvectors of $\mathbf{A}_0$ and $\mathbf{C}_0$ is the diagonal matrix with entries $(C_1, \ldots, C_N)$, inverses of the eigenvalues of $\mathbf{A}_0$. Entries of $\mathbf{U}$ are the turn ratios of the multiport Belevitch transformer corresponding to the purely capacitive stage in Fig. 16. In the case of no direct electrostatic interaction between the qubit port terminals, $\mathbf{U}$ will be simply the identity matrix.

For the internal modes of frequency $\omega_{R_k} = 1/(L_{R_k}C_{R_k})^{1/2}$, we choose a characteristic impedance of $Z_0 = 1/\omega_{R_k}$ that will make all $C_{R_k} = 1$ for $1 \leq k \leq M$. There is a freedom in the choice of this characteristic impedance; this choice should have no effect on the physical coupling rates. With this choice, we have $L_{R_k} = 1/\omega_{R_k}^2$. Then, with $\mathbf{A}_k$ values being rank-1 matrices and with our choice of $C_{R_k} = 1$

$$\mathbf{A}_k = r_k^T r_k \tag{66}$$

where r_k is the row-vector $r_k = (r_{k1}, \ldots, r_{kN})$ for $1 \leq k \leq M$. r_k values constitute rows of turn ratios of the multiport Belevitch transformer matrix $\mathbf{R}$ connecting the internal modes to the ports.

The final purely inductive stage can be synthesized in a similar way to the purely capacitive dc stage with an eigendecomposition of the $\mathbf{A}_\infty$ matrix

$$\mathbf{A}_\infty = \mathbf{T}^T \mathbf{L}_\infty \mathbf{T} \tag{67}$$

with $\mathbf{T}$ being the orthonormal matrix holding the eigenvectors and the diagonal matrix $\mathbf{L}_\infty$ holding the inductances $(L_1^\infty, \ldots, L_N^\infty)$.

Using the lumped element circuit quantization method in [19] together with a technique to handle multiport Belevitch transformers [27], we can identify the degrees of freedom in the Cauer circuit in Fig. 16 and write an equation of motion. The effective fundamental loop matrix defined in [19, eq. (21)] is

$$\mathbf{F}_C = \begin{pmatrix} \mathbf{U} \\ -\mathbf{R}\mathbf{U} \end{pmatrix}. \tag{68}$$

The Hamiltonian is

$$\mathcal{H} = \frac{1}{2}\mathbf{Q}^T\mathbf{C}^{-1}\mathbf{Q} + \frac{1}{2}\mathbf{\Phi}^T\mathbf{M}_0\mathbf{\Phi} - \sum_{i=1}^{N} E_{J_i}\cos(\varphi_{J_i}) \tag{69}$$

where $\mathbf{\Phi} = (\Phi_{J_1}, \ldots, \Phi_{J_N}, \Phi_{R_1}, \ldots, \Phi_{R_M})^T$ being the flux coordinate vector. φ_{J_i} is the phase of the junction i related to

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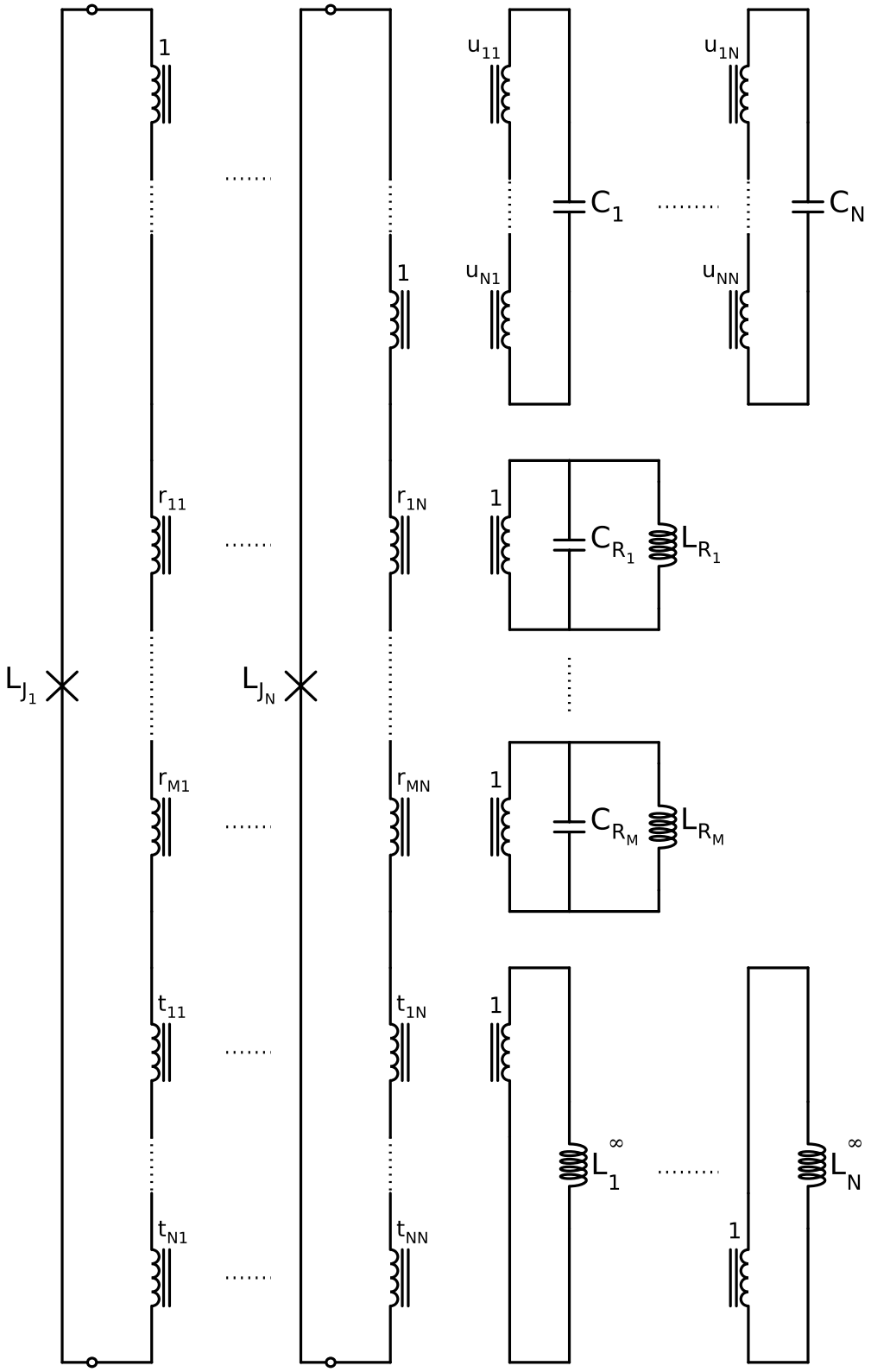


Fig. 16. Multiport canonical Cauer circuit shunted with Josephson junctions. On the left, we have N "qubit ports" shunted by the Josephson junctions J_i. Port terminals are shown with open dots. On the top right, we have the purely capacitive stage providing the total qubit shunt capacitances C_i. This stage is coupled to the rest of the circuit with the multiport Belevitch transformer $\mathbf{U}$. LC tank circuits on the middle right correspond to the internal modes. Interactions between the qubit ports and the internal modes are mediated by the multiport Belevitch transformer of turn ratio matrix $\mathbf{R}$. On the bottom right, we have the purely inductive stage corresponding to the pole at infinity of the impedance expansion in (16) consisting of linear inductors $L_1^{\infty}, \ldots, L_N^{\infty}$. This stage can safely be neglected since, in most of the physical situations, every inductor will have a parasitic capacitance.

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the flux across it by the Josephson relation $\Phi_{J_i} = (\Phi_0/2\pi)\varphi_{J_i}$, for $1 \le i \le N$. Φ_{R_k} is the flux across the inductor of the internal mode k, $1 \le k \le M$. E_{J_i} is the Josephson energy of junction i related to its inductance L_{J_i} by $E_{J_i} = (\Phi_0/2\pi)^2(1/L_{J_i})$. The capacitance matrix $\mathbf{C}$ is given by

$$\mathbf{C} = \mathbf{F}_C\mathbf{C}_0\mathbf{F}_C^T + \mathbf{C}_R = \begin{pmatrix} \mathbf{U}\mathbf{C}_0\mathbf{U}^T & -\mathbf{U}\mathbf{C}_0\mathbf{U}^T\mathbf{R}^T \\ -\mathbf{R}\mathbf{U}\mathbf{C}_0\mathbf{U}^T & \mathbf{C}_R + \mathbf{R}\mathbf{U}\mathbf{C}_0\mathbf{U}^T\mathbf{R}^T \end{pmatrix}. \quad (70)$$

The capacitance matrix becomes

$$\mathbf{C} = \begin{pmatrix} \mathbf{C}_0 & -\mathbf{C}_0\mathbf{R}^T \\ -\mathbf{R}\mathbf{C}_0 & \mathbf{1}_{M\times M} + \mathbf{R}\mathbf{C}_0\mathbf{R}^T \end{pmatrix} \quad (71)$$

in the absence of direct electrostatic dipole interactions between the ports since $\mathbf{U}$ is the identity matrix in that case and with our choice of $\mathbf{C}_R = \mathbf{1}_{M\times M}$ for the capacitances of the internal modes.

$\mathbf{M}_0$ is the diagonal matrix holding the inverses of the inductances of the internal modes on its diagonal

$$\mathbf{M}_0 = \begin{pmatrix} \mathbf{0}_{N\times N} & & & \mathbf{0} \\ & 1/L_{R_1} & & \\ & & \ddots & \\ \mathbf{0} & & & 1/L_{R_M} \end{pmatrix}. \quad (72)$$

Appendix B
Derivation of the Coupling Rates of the Qubits to the Voltage Drives

In this section, we augment the canonical Cauer circuit in Fig. 16 by including the drive lines, as shown in Fig. 17. We added N_D drive lines, hence N_D more ports. Drive line for the qubit i consists of the voltage source $V_{d(i)}$ driving the transmission line of characteristic impedance Z_0 whose other end is connected to the drive port $p(d(i))$ [here, we are assuming that $d(i)$ is the index number of the voltage source corresponding to the qubit i]. Synthesis of such a circuit from an impedance matrix $\mathbf{Z}(\omega)$ proceeds, as described in the previous section; this time with $N + N_D$ ports.

Again using the method in [19], we obtain the following Hamiltonian for the augmented Cauer circuit in Fig. 17:

$$\mathcal{H} = \frac{1}{2}(\mathbf{Q} - \mathbf{C}_Q * \mathbf{V}_V)^T\mathbf{C}^{-1}(\mathbf{Q} - \mathbf{C}_Q * \mathbf{V}_V) + U(\mathbf{\Phi}) \quad (73)$$

with the potential energy

$$U(\mathbf{\Phi}) = \frac{1}{2}\mathbf{\Phi}^T\mathbf{M}_0\mathbf{\Phi} - \sum_{i=1}^{N} E_{J_i}\cos(\varphi_{J_i}) \quad (74)$$

and as in the previous section, $\mathbf{\Phi} = (\Phi_{J_1}, \ldots, \Phi_{J_N}, \Phi_{R_1}, \ldots, \Phi_{R_M})^T$ is the flux coordinate vector. The capacitance matrix $\mathbf{C}$ is given by

$$\mathbf{C} = \mathbf{F}_C\mathbf{C}_S\mathbf{F}_C^T + \mathbf{C}_R \quad (75)$$

where $\mathbf{C}_S$ is the diagonal matrix holding the total shunt capacitances seen at the ports

$$\mathbf{C}_S = \begin{pmatrix} \mathbf{C}_0 & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_D \end{pmatrix} \quad (76)$$

where $\mathbf{C}_0$ and $\mathbf{C}_D$ are $N \times N$ and $N_D \times N_D$ diagonal matrices holding total capacitances shunting the qubit and drive ports, respectively, such that

$$\mathbf{C}_0 = \begin{pmatrix} C_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & C_N \end{pmatrix} \quad (77)$$

$$\mathbf{C}_D = \begin{pmatrix} C_{N+1} & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & C_{N+N_D} \end{pmatrix} \quad (78)$$

where capacitances $(C_1, \ldots, C_N, C_{N+1}, \ldots, C_{N+N_D})$ are shown in Fig. 17 in the purely capacitive stage coupled to the rest of the circuit with the multiport Belevitch transformer $\mathbf{U}$. $\mathbf{C}_R$ is again the identity matrix in the resonator subspace, and neglecting any electrostatic dipole interaction between the ports (i.e., $\mathbf{U} = \mathbf{1}$), the fundamental loop matrix $\mathbf{F}_C$ is given by

$$\mathbf{F}_C = \begin{pmatrix} \mathbf{1}_{N\times N} & \mathbf{0}_{N\times N_D} \\ -\mathbf{R} & -\mathbf{V} \end{pmatrix} \quad (79)$$

where $\mathbf{F}_C$, $\mathbf{R}$, and $\mathbf{V}$ are $(N + M) \times (N + N_D)$, $(M \times N)$, and $(M \times N_D)$ matrices, respectively. Matrices $\mathbf{R}$ and $\mathbf{V}$ are multiport Belevitch transformer matrices (with turn ratio entries r_{ki} and v_{kd} as shown in Fig. 17 for $1 \le k \le M$, $1 \le i \le N$, and $1 \le d \le N_D$) mediating the couplings of the internal modes to the qubits and the voltage sources, respectively. The diagonal matrix $\mathbf{M}_0$ again holds the inverses of the inductances of the internal modes on its diagonal

$$\mathbf{M}_0 = \begin{pmatrix} \mathbf{0}_{N\times N} & & & \mathbf{0} \\ & 1/L_{R_1} & & \\ & & \ddots & \\ \mathbf{0} & & & 1/L_{R_M} \end{pmatrix}. \quad (80)$$

$\mathbf{V}_V = (V_1, \ldots, V_{N_D})$ is the vector of voltage sources and $*$ is the time-convolution operator. $\mathbf{C}_Q$ is the $(N + M) \times N_D$ matrix coupling the voltage source $\mathbf{V}_V$ vector to the charge coordinates $\mathbf{Q}$ and is given by

$$\mathbf{C}_Q = \mathbf{C}_V + \mathcal{C}_V. \quad (81)$$

As we will show in the following, $\mathbf{C}_V$ is frequency-independent, whereas $\mathcal{C}_V$ is nonzero only for ac voltage drives. $\mathbf{C}_V$ is given in [19, eq. (23)] as:

$$\mathbf{C}_V = \mathbf{F}_C\mathbf{C}_S\mathbf{F}_{VC}^T \quad (82)$$

where the loop matrix $\mathbf{F}_{VC}$ is given by

$$\mathbf{F}_{VC} = \begin{pmatrix} \mathbf{0}_{N\times N} & \mathbf{1}_{N\times N_D} \end{pmatrix}. \quad (83)$$

$\mathcal{C}_V(\omega)$ is given in [27, eq. (7.25)], which is an extension of [19, eq. (23)] to ac voltage sources

$$\mathcal{C}_V(\omega) = \bar{\mathbf{m}}\bar{\mathbf{C}}_Z\bar{\mathbf{m}}_V^T \quad (84)$$

where from [27, eqs. (7.19)–(7.21)]

$$\bar{\mathbf{m}} = \mathbf{F}_C\mathbf{C}_S\mathbf{F}_{ZC}^T \quad (85)$$

$$\bar{\mathbf{m}}_V = \mathbf{F}_{VC}\mathbf{C}_S\mathbf{F}_{ZC}^T \quad (86)$$

$$\bar{\mathbf{C}}_Z(\omega) = -i\omega\mathbf{Z}_0\left[\mathbf{1} + i\omega\mathbf{F}_{ZC}\mathbf{C}_S\mathbf{F}_{ZC}^T\mathbf{Z}_0\right]^{-1}. \quad (87)$$

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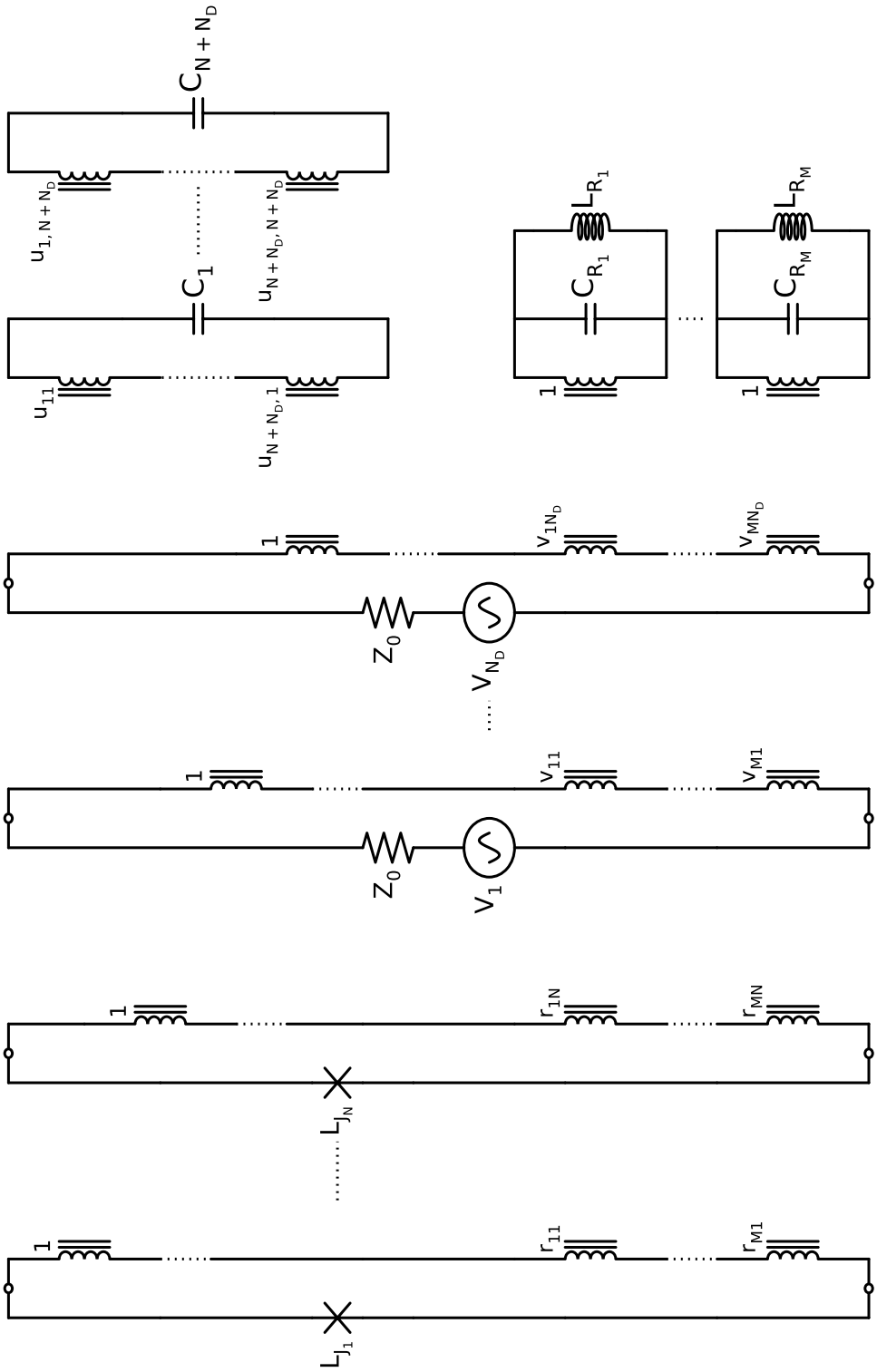


Fig. 17. Canonical Cauer circuit in Fig. 16 with the purely inductive stage dropped and augmented with N_D control/readout lines. Drive lines are modeled with transmission lines of infinite extent such that they are represented by constant characteristic impedances Z_0 shunted with the voltage sources V_d for $1 \leq d \leq N_D$. Coupling of the drive lines to the internal modes is mediated by the multiport Belevitch tranformer $\mathbf{V}$ with entries v_{kd} for $1 \leq k \leq M$ and $1 \leq d \leq N_D$. The purely capacitive stage has now N_D additional capacitances $C_{N+1}, \ldots, C_{N+N_D}$ corresponding to the shunt capacitances of the drive ports (we define the $N_D \times N_D$ diagonal matrix $\mathbf{C}_D$ in (78) holding total shunt capacitances of the drive ports). The $(N+N_D) \times (N+N_D)$ multiport transformer $\mathbf{U}$ with entries couples the shunt capacitances to the qubit and drive ports.

Here, $\mathbf{F}_{ZC} = \mathbf{F}_{VC}$ given in (83) and $\mathbf{Z}_0$ is the $N_D \times N_D$ matrix giving the multiport impedance seen at the drive ports looking into the environment away from the chip and is simply the diagonal matrix consisting of diagonal entries Z_0.

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We observe that $\mathbf{C}_V = \bar{\mathbf{m}}$ since $\mathbf{F}_{VC} = \mathbf{F}_{ZC}$. Noting

$$\mathbf{F}_{ZC}\mathbf{C}_S\mathbf{F}_{ZC}^T = \mathbf{C}_D \tag{88}$$

we write

$$\bar{\mathbf{C}}_Z(\omega) = -i\omega\mathbf{Z}_0[\mathbf{1} + i\omega\mathbf{C}_D\mathbf{Z}_0]^{-1}. \tag{89}$$

We now work out $\bar{\mathbf{m}}$ using (76), (79), (83), and (85) and noting $\mathbf{F}_{ZC} = \mathbf{F}_{VC}$

$$\begin{aligned}\bar{\mathbf{m}} &= \mathbf{F}_C\mathbf{C}_S\mathbf{F}_{ZC}^T \\ &= \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ -\mathbf{R} & -\mathbf{V} \end{pmatrix}\begin{pmatrix} \mathbf{C}_0 & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_D \end{pmatrix}(\mathbf{0}\ \mathbf{1})^T \\ &= \begin{pmatrix} \mathbf{0} \\ -\mathbf{V}\mathbf{C}_D \end{pmatrix}.\end{aligned} \tag{90}$$

Applying the capacitance rescaling $\mathbf{\Phi}_J \rightarrow \mathbf{C}_0^{1/2}\mathbf{\Phi}_J$ and the transformation in (21) to $\bar{\mathbf{m}}$ in (90)

$$\bar{\mathbf{m}} \rightarrow \mathbf{T}^t\begin{pmatrix} \mathbf{C}_0^{-1/2} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{pmatrix}\bar{\mathbf{m}} \tag{91}$$

we get

$$\bar{\mathbf{m}} = \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{R}\mathbf{C}_0^{1/2} & \mathbf{1} \end{pmatrix}\begin{pmatrix} \mathbf{C}_0^{-1/2} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{pmatrix}\begin{pmatrix} \mathbf{0} \\ -\mathbf{V} \end{pmatrix}\mathbf{C}_D = \begin{pmatrix} \mathbf{0} \\ -\mathbf{V}\mathbf{C}_D \end{pmatrix}. \tag{92}$$

We note that $\bar{\mathbf{m}}$ is unaffected by this transformation. Since $\mathbf{C}_V = \bar{\mathbf{m}}$, we have after the transformations

$$\mathbf{C}_V = \begin{pmatrix} \mathbf{0} \\ -\mathbf{V}\mathbf{C}_D \end{pmatrix}. \tag{93}$$

Noting

$$\bar{\mathbf{m}}_V = \mathbf{F}_{VC}\mathbf{C}\mathbf{F}_{ZC}^T = \mathbf{F}_{ZC}\mathbf{C}\mathbf{F}_{ZC}^T = \mathbf{C}_D \tag{94}$$

we can write (84) as

$$\mathcal{C}_V(\omega) = \bar{\mathbf{m}}\bar{\mathbf{C}}_Z\bar{\mathbf{m}}_V^T = \begin{pmatrix} \mathbf{0} \\ \mathbf{V} \end{pmatrix}\widetilde{\mathbf{C}}_Z(\omega) \tag{95}$$

where we defined the $N_D \times N_D$ matrix

$$\widetilde{\mathbf{C}}_Z(\omega) = i\omega\mathbf{C}_D\mathbf{Z}_0[\mathbf{1} + i\omega\mathbf{C}_D\mathbf{Z}_0]^{-1}\mathbf{C}_D. \tag{96}$$

We have one final step to do, that is, to apply the Schrieffer–Wolff transformation to $\mathbf{C}_V$ and $\mathcal{C}_V(\omega)$, such that

$$\mathbf{C}_V \rightarrow \exp(-\mathbf{S})\mathbf{C}_V \tag{97}$$

$$\mathcal{C}_V(\omega) \rightarrow \exp(-\mathbf{S})\mathcal{C}_V(\omega). \tag{98}$$

Using [29, eqs. (B.4) and (B.12a)] and noting block structures of matrices $\mathbf{S}$, C_V, and $\mathcal{C}_V(\omega)$, we first define the following $(N+M) \times N_D$ matrix $\mathbf{D}$ having the (i,d)th entry D_{id} in the qubit subspace:

$$\begin{aligned}D_{id} &= \left[\exp(-\mathbf{S})\begin{pmatrix} \mathbf{0} \\ -\mathbf{V} \end{pmatrix}\right]_{id} \\ &= -\sum_k(\mathbf{M}_1)_{ik}\frac{v_{kd}}{\omega_i^2 - \omega_{R_k}^2} \\ &= -\omega_i^2 C_i^{1/2}\sum_k\frac{r_{ki}v_{kd}}{\omega_i^2 - \omega_{R_k}^2} \\ &= -\omega_i^2 C_i^{1/2}\sum_k\frac{[\mathbf{A}_k]_{i,p(d)}}{\omega_i^2 - \omega_{R_k}^2} \\ &= \omega_i C_i^{1/2}\mathrm{Im}[Z_{i,p(d)}(\omega_i)] \\ &= \mathrm{Im}[Z_{i,p(d)}(\omega_i)]/\sqrt{L_i}\end{aligned} \tag{99}$$

where v_{kd} is the (k,d)th entry of $\mathbf{V}$ for $1 \le k \le M$ and $1 \le d \le N_D$. In the third line in (99), we used (28) to replace $(\mathbf{M}_1)_{ik}$ with $\omega_i^2 C_i^{1/2} r_{ki}$, and in the fourth line, $[\mathbf{A}_k]_{i,p(d)} = r_{ki}v_{kd}$ where $[\mathbf{A}_k]_{i,p(d)}$ is the entry of the residue matrix $\mathbf{A}_k$ in the impedance expansion in (16) for the circuit in Fig. 17 connecting the qubit port i to drive port [with port index $p(d)$] corresponding to the voltage source V_d. Hence, $\mathbf{C}_Q$ transforms to

$$\mathbf{C}_Q \rightarrow \mathbf{C}_q = \mathbf{D}(\mathbf{C}_D - \widetilde{\mathbf{C}}_Z(\omega)). \tag{100}$$

Then, one can write the following Hamiltonian in the final frame corresponding to $\widetilde{\mathbf{M}}_1$ in (24):

$$H = \frac{1}{2}(\boldsymbol{q} - \mathbf{C}_q * \mathbf{V}_V)^T(\boldsymbol{q} - \mathbf{C}_q * \mathbf{V}_V) + \frac{1}{2}\boldsymbol{\phi}^T\widetilde{\mathbf{M}}_1\boldsymbol{\phi} + \mathcal{O}(\boldsymbol{\varphi}_J^4) \tag{101}$$

with the $(N+M) \times N_D$ matrix $\mathbf{C}_q$ giving the couplings of the voltage drives $\mathbf{V}_V$ to the momentum degrees of freedom $\boldsymbol{q}$ in the final frame. After quantization by introducing $\hat{q}_i = -i(\hbar/2Z_i)^{1/2}(\hat{b}_i - \hat{b}_i^\dagger)$ and noting that the characteristic impedance Z_i of the qubit mode i is $Z_i = 1/\omega_i$, we get the drive term on qubit i due to voltage source V_d

$$H_{id}^D = i\sqrt{\frac{\hbar\omega_i}{2}}[\mathbf{C}_q(\omega_d)]_{i,d}V_d(\hat{b}_i - \hat{b}_i^\dagger) \tag{102}$$

where $[\mathbf{C}_q(\omega_d)]_{i,d}$ is the (i,d)th entry of $\mathbf{C}_q$ evaluated at the frequency ω_d (we assumed that V_d is a single-tone sinusoidal voltage drive at frequency ω_d). In the case of zero off-chip crosstalk, $\widetilde{\mathbf{C}}_Z$ is diagonal and using (99) and (100), we have

$$H_{id}^D = i\sqrt{\frac{\hbar\omega_i}{2L_i}}\mathrm{Im}[Z_{i,p(d)}(\omega_i)]\frac{C_{p(d)}V_d(\hat{b}_i - \hat{b}_i^\dagger)}{1 + i\omega_d Z_0 C_{p(d)}} \tag{103}$$

where $C_{p(d)}$ is the dth diagonal entry of $\mathbf{C}_D$. Here, we note that ε_{id} in (7) is

$$\varepsilon_{id} = i\sqrt{\frac{\omega_i}{2\hbar L_i}}\mathrm{Im}[Z_{i,p(d)}(\omega_i)]\left(\frac{C_{p(d)}}{1 + i\omega_d Z_0 C_{p(d)}}\right) \tag{104}$$

which can also be written as

$$\varepsilon_{id} = \sqrt{\frac{\omega_i}{2\hbar L_i}}\mathrm{Im}[Z_{i,p(d)}(\omega_i)]\frac{e^{i\theta_d}C_{p(d)}}{\sqrt{1 + \omega_d^2 Z_0^2 C_{p(d)}^2}} \tag{105}$$

with $\theta_d = (\pi/2) - \arctan(\omega_d Z_0 C_{p(d)})$.

One can then define the following quantity (in units of dB) as a measure of classical on-chip crosstalk on qubit i while driving qubit j:

$$X_{ij} = 20\log_{10}\left(\left|\frac{\mathrm{Im}[Z_{i,d(j)}(\omega_i)]}{\mathrm{Im}[Z_{j,d(j)}(\omega_j)]}\right|\right). \tag{106}$$

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TABLE II

TABLE TO COMPARE EXPERIMENTAL MEASUREMENTS WITH THE THEORETICAL J-COUPLING PREDICTIONS. THE AVERAGE PREDICTION ERROR IS 4.5%. SIMULATIONS HAVE CONVERGED WITH LESS THAN 1% ERROR FOR THE OFF-DIAGONAL IMPEDANCES

Bus	f_b (GHz)	ZZ (KHz)	$J^{(M)}$ (MHz)	$J^{(S)}$ (MHz)	Unit Cell	Error
(1,2)	6.25	75 (74)	2.12	2.37	A	11.8 %
(2,3)	6.45	96 (95)	1.96	1.93	B	1.5 %
(3,4)	6.25	37 (38)	1.55	1.64	A	5.8 %
(4,5)	6.45	60 (62)	1.67	1.72	B	3.0 %
(5,6)	6.25	78 (76)	1.99	2.11	A	6.0 %
(6,7)	6.45	55 (60)	2.01	2.11	B	5.0 %
(7,8)	6.25	58 (59)	1.85	1.91	A	3.2 %
(9,10)	6.25	26 (25)	1.62	1.72	A	6.2 %
(10,11)	6.45	40 (43)	1.66	1.70	B	2.4 %
(11,12)	6.25	38 (42)	1.58	1.61	A	1.9 %
(12,13)	6.45	51 (50)	1.57	1.59	B	1.3 %
(13,14)	6.25	43 (46)	1.50	1.52	A	1.3 %
(14,15)	6.45	74 (69)	1.55	1.56	B	0.6 %
(15,16)	6.25	103 (106)	1.84	1.94	A	5.4 %
(1,16)	6.65	46 (43)	1.58	1.74	A	10.1 %
(2,15)	6.65	53 (55)	1.62	1.51	A	6.8 %
(3,14)	6.65	51 (50)	1.33	1.37	A	3.0 %
(4,13)	6.65	27 (27)	1.30	1.31	A	0.8 %
(5,12)	6.65	54 (51)	1.36	1.34	A	1.5 %
(6,11)	6.65	66 (68)	1.63	1.55	A	4.9 %
(7,10)	6.65	69 (71)	1.62	1.44	A	11.1 %
(8,9)	6.65	26 (25)	1.36	1.42	A	4.4 %
(2,15)	6.65	53 (55)	1.62	1.55	B	4.3 %
(3,14)	6.65	51 (50)	1.33	1.25	B	6.0 %
(4,13)	6.65	27 (27)	1.30	1.32	B	1.5 %
(5,12)	6.65	54 (51)	1.36	1.35	B	0.7 %
(6,11)	6.65	66 (68)	1.63	1.55	B	4.9 %
(7,10)	6.65	69 (71)	1.62	1.47	B	9.3 %
Avg.						4.5 %

APPENDIX C
DETAILS OF THE SIMULATIONS FOR THE EXPERIMENTAL VALIDATION OF THE EXCHANGE COUPLING IMPEDANCE FORMULA

ZZ-rates between all pairs of qubits have been experimentally measured and listed in the third column of Table II. The formula in [30, eq. (58)] relates the ZZ-rates to J-coupling rates as follows:

$$\zeta_{ij} = (E_{11} - E_{10} - E_{01})/\hbar = -\frac{2J_{ij}^2(\delta_i + \delta_j)}{(\Delta_{ij} + \delta_i)(\delta_j - \Delta_{ij})} \quad (107)$$

where ζ_{ij} is the ZZ-rate, $\Delta_{ij} = \omega_i - \omega_j$ is the detuning, and J_{ij} is the exchange coupling rate between qubit i and j. δ_i and δ_j are the anharmonicities of qubits i and j. E_{10} and E_{01} are the energies when one of the qubits is in its excited state and E_{11} is the energy when both qubits are in their excited state. Using this formula, we back up the J-coupling rates from the experimental ZZ measurements between all qubit pairs that are connected by a bus and list them in the fourth column [with $J^{(M)}$ header] of Table II. We compare these values to the values we obtain from the application of the formula in (5) using the experimentally measured values ω_i and ω_j for the qubit frequencies and anharmonicities δ_i and δ_j. To compute the linear qubit inductances L_i and L_j, we use the following relations that we derived in Section E:

$$\omega_i^2 = \omega_{J_i}^2 \left(1 - \frac{2E_C^{(i)}}{\hbar\omega_i}\right) \quad (108)$$

$$\delta_i = -E_C^{(i)} \left(\frac{\omega_{J_i}}{\omega_i}\right)^2 \quad (109)$$

to solve for the charging energy $E_C^{(i)}$ of the qubit i as

$$E_C^{(i)} = -\frac{\delta_i}{1 - \frac{2\delta_i}{\omega_i}}. \quad (110)$$

One can then determine C_i and L_i from

$$\omega_i = 1/\sqrt{L_i C_i} \quad (111)$$

$$E_C^{(i)} = \frac{e^2}{2C_i}. \quad (112)$$

We computed the off-diagonal impedance entries Z_{ij} appearing in (5) by performing high-accuracy full-wave FEM simulations in Ansys HFSS at each experimentally measured qubit frequency (see the following and Fig. 18). The values obtained from the application of the formula in (5) are listed in the fifth column [with header $J^{(S)}$] of Table II.

In the first column in Table II, we label buses by the indices (i, j) of the qubit pair that the corresponding bus connects. The second column lists the bus frequencies. In the third column, we list experimentally measured ZZ-rates between the qubits connected by the corresponding bus. The numbers in parenthesis in the third column correspond to the measured ZZ rates with the roles of the qubits switched in the measurement protocol. In principle, these numbers should be the same. We attribute the discrepancy (up to 10%) to the noise present in the chip. The fourth column with

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header $J^{(M)}$ lists the J-coupling rates we back up from the experimental ZZ-rate values in column 3 using (107). In the fifth column with header $J^{(S)}$, we list the J-coupling rates we calculated by simulating the impedance entries Z_{ij} at each qubit frequency as we describe in Fig. 18 and in the next paragraph and applying the formula in (5). In the sixth column, we list the type of unit cell the bus connecting the qubit pair (i, j) belongs to (note that the vertical buses—except the ones at left and right ends—in Fig. 6 belong to both types of unit cell), and in the last column, we list the errors between the J-coupling rates we calculate from the experimental ZZ-rate measurements [column 4—header $J^{(M)}$] and our theoretical prediction applying the formula in (5) [column 5—header $J^{(S)}$]. The average error is 4.5% for our prediction for the J-couplings using the formula in (5) and our simulations. We note that this is given the variations up to 10% in the ZZ-rate measurements and less than 1% error in our simulations for the off-diagonal impedance entries Z_{ij}. We also note that the largest errors occur between the qubits that are on the edges of the chip.

To keep accuracy at the levels we needed with the memory resources available, we simulated only 2×2 unit cells of the 2×8 16 qubit device, as shown in Fig. 18. There are two types of unit cells: A and B. We defined lumped ports at the locations of the Josephson junctions and simulated the multiport impedance matrix between these ports at each qubit frequency. Fig. 18(a), (c), and(e) corresponds to unit cell A's 3-D view, top view, and top view with the FEM meshing, respectively. Similarly, Fig. 18(b), (d), and (f) corresponds to unit cell B.

Appendix D
Defining the Qubit Ports and the Drive Ports in the 3-D Finite-Element Electromagnetic Simulators

In the main text, we described in words how to define the qubit ports and the drive ports. In this section, we illustrate the definition of the ports with the help of the 3-D model of a 7-Qubit Device in Ansys HFSS, as shown in Fig. 19 (HFSS is a high-frequency finite-element electromagnetics simulator). The device consists of a quantum chip (shown in light blue in the middle) packaged together with a PCB supporting transmission lines carrying the drive and readout signals to/from the chip. The metallization of the PCB is shown in orange and the dielectric of the PCB is shown in burgundy color in Fig. 19. For the definition of the drive ports, we choose a bounding box enclosing the quantum chip and some part of the PCB. The boundaries of the box should be chosen far enough from the chip. As we stated in the main text, the exact position of this boundary can be determined with a TDR experiment/simulation. Ideally, we would like to put the boundary at the location where signals traveling in the transmission lines of the PCB start to see a change in the constant impedance of the transmission lines. This happens where the signals enter the discontinuity region between the PCB and the chip. The drive ports are usually defined as wave ports in HFSS to which it is assumed that a constant impedance transmission line is connected. An example of a drive port is shown in Fig. 19(d) as the magenta rectangle on one of the side surfaces of the bounding box shown in Fig. 19(b).

Qubit ports are defined as lumped ports in HFSS. This is shown in Fig. 19(e) and (f). The qubit port is the small magenta square shown in Fig. 19(f). HFSS puts a differential excitation between the edges of this square touching the junction terminals.

Appendix E
Expansion of the Junction Potentials

Qubit anharmonicities and dispersive shifts between the modes are obtained after including the nonlinear terms in the junction potentials. For this, we use the normal ordered expansion as given in [25, eq. (16)]

$$H = H_0 + H_\gamma + H_\beta + \mathcal{O}\big(\hat{\varphi}_J^6\big) \tag{113}$$

with

$$H_\gamma = -\sum_{pp'} \gamma_{pp'} \big(2\hat{a}_p^\dagger a_{p'} + \hat{a}_p \hat{a}_{p'} + \hat{a}_p^\dagger \hat{a}_{p'}^\dagger\big) \tag{114}$$

$$H_\beta = -\sum_{pp'qq'} \beta_{pp'qq'} \big(6\hat{a}_p^\dagger \hat{a}_{p'}^\dagger \hat{a}_q \hat{a}_{q'} + 4\hat{a}_p^\dagger \hat{a}_{p'}^\dagger \hat{a}_q^\dagger \hat{a}_{q'} + 4\hat{a}_p^\dagger \hat{a}_{p'} \hat{a}_q \hat{a}_{q'} + \hat{a}_p \hat{a}_{p'} \hat{a}_q \hat{a}_{q'} + \hat{a}_p^\dagger \hat{a}_{p'}^\dagger \hat{a}_q^\dagger \hat{a}_{q'}^\dagger\big) \tag{115}$$

where H_0 is the linearized part of the Hamiltonian obtained after replacing the junctions with linear inductors, p, p', q, and q' are the labels of the harmonic modes in the basis defined by H_0, and $\hat{a}_p(\hat{a}_p^\dagger)$ is the annihilation (creation) operator of mode p. This expansion was originally done in [25] in a diagonal frame, whereas here we will expand in the block-diagonalized frame corresponding to the matrix $\widetilde{\mathbf{M}}_1$ in (24). That is, the linearized Hamiltonian H_0 in our case is the linear part of the Hamiltonian given in (25)

$$H_0 = \frac{1}{2}\boldsymbol{q}^T\boldsymbol{q} + \frac{1}{2}\boldsymbol{\phi}^T\widetilde{\mathbf{M}}_1\boldsymbol{\phi}. \tag{116}$$

In that frame, the capacitance matrix is unity and the coordinate vector holds the flux variables $\boldsymbol{\phi} = (\phi_1 \ldots \phi_{N+M})$. The first N coordinates correspond to qubit modes and the last M coordinates correspond to the resonator modes (or internal modes). $\boldsymbol{q}$ is the vector of momenta conjugate to coordinates $\boldsymbol{\phi}$. The flux operators $\boldsymbol{\phi}$ of the modes in the final frame can be related to the fluxes $\boldsymbol{\Phi}$ in the initial frame by the total coordinate transformation

$$\boldsymbol{\Phi} = \begin{pmatrix} \boldsymbol{\Phi}_J \\ \boldsymbol{\Phi}_R \end{pmatrix} = \begin{pmatrix} \mathbf{C}_0^{-1/2} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{pmatrix} \boldsymbol{\alpha}\boldsymbol{\phi} \tag{117}$$

where $\boldsymbol{\alpha} = \mathbf{T}\exp(\mathbf{S})$; matrices $\mathbf{C}_0$, $\mathbf{T}$, and $\mathbf{S}$ are defined in the text in (18), (21), and (24), respectively. In particular

$$\boldsymbol{\Phi}_J = \begin{pmatrix} \mathbf{C}_0^{-1/2} & \mathbf{0}_{N\times M} \end{pmatrix} \boldsymbol{\alpha}\boldsymbol{\phi} \tag{118}$$

where $\boldsymbol{\Phi}_J = (\Phi_{J_1} \ldots \Phi_{J_N})^T$ is the vector of fluxes across the Josephson junctions. Hence,

$$\Phi_{J_i} = \frac{1}{\sqrt{C_i}} \left[\sum_{j=1}^{N} \alpha_{ij}\phi_j + \sum_{k=1}^{M} \alpha_{i,k+N}\phi_{k+N} \right] \tag{119}$$

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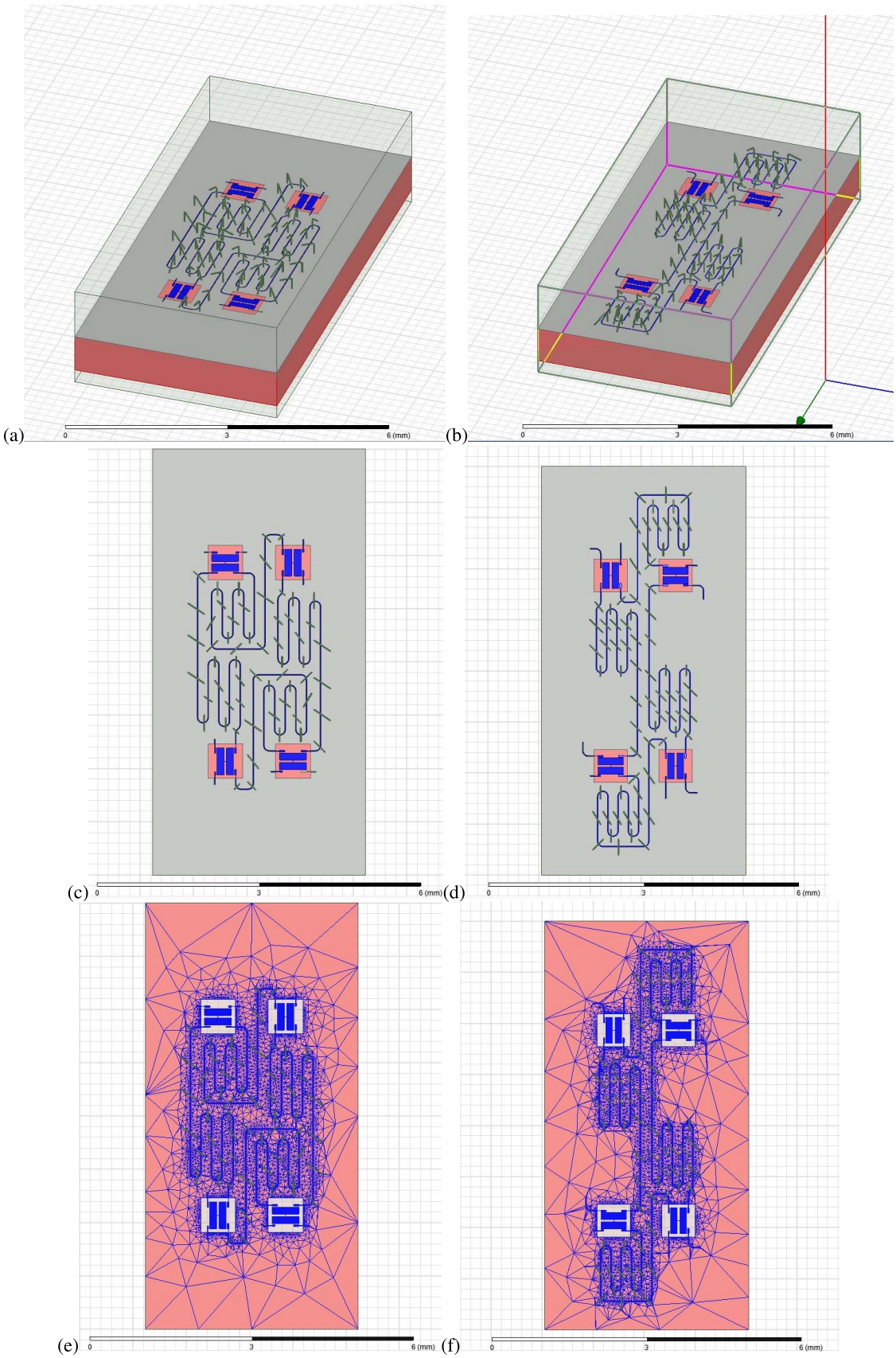


Fig. 18. Screen captures of Ansys HFSS(r) 3-D FEM electromagnetic simulation geometries of two different unit cells: A (left) and B (right). Off-diagonal impedance entries Z_{ij} converged with an error below 1% in our simulations. (a) and (b) 3-D views of the unit cells A, B. (c) and (d) Top views of the unit cells A, B. (e) and (f) Top views of the units cells A and B with the meshes displayed.

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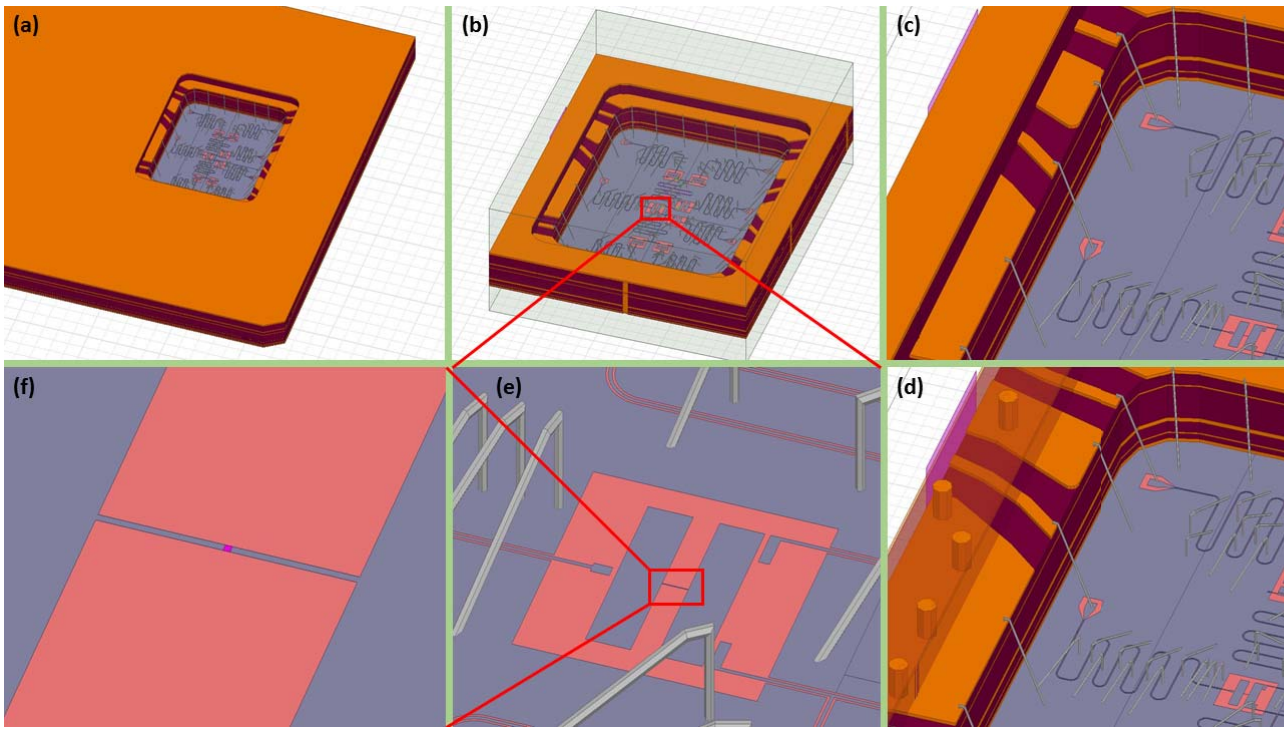


Fig. 19. (a) HFSS model of a 7-Qubit Device to illustrate the definition of the drive ports and the qubit ports. The device consists of a superconducting quantum chip packaged together with a PCB. Light blue region in the middle is the chip metallization. In orange is the copper metallization and in burgundy color is the dielectric of the PCB. (b) Bounding box is defined over the side surfaces of which the drive ports are defined as rectangles, see the rectangular area in magenta in (d). These rectangles should be chosen large enough to enclose all the fields due to the drive excitations. (c) Close-up view of the drive line of one of the qubits. The copper central trace of the drive line in the PCB is capacitively coupled to the on-chip readout resonator (the meandered structures) with a wirebond and a launchpad. (d) Same picture in (c) with parts of the PCB made transparent to make the drive port visible (magenta rectangle on the side surface of the bounding box). The drive port is usually defined as a wave port (to which it is assumed that a constant impedance transmission line is connected) and the size of the rectangle should be chosen properly to enclose the fields due to the excitation at the port. (e) Close-up view of qubit pads (two identical rectangles in light blue). Light gray are wirebonds and the pink region in the qubit pocket is the upper surface of the substrate underneath the chip metallization. (f) Qubit port is the small square shown in magenta defined between the leads connecting the qubit pads (large light blue regions) to the Josephson junction (not shown).

where indices i and j label qubit modes and k labels resonator modes with $1 \leq i, j \leq N$ and $1 \leq k \leq M$. The $(N+M) \times (N+M)$ matrix $\boldsymbol{\alpha}$ has the entries

$$\alpha_{ii} = 1 - \mathrm{Im}\left[Z_{ii}^{(ac)}(\omega_i)\right]/Z_i \tag{120}$$

$$\alpha_{ij} = -\mathrm{Im}[Z_{ij}(\omega_i)]/Z_i \tag{121}$$

$$\alpha_{i,k+N} = r_{ki} C_i^{1/2} \left(\frac{\omega_{R_k}^2}{\omega_{R_k}^2 - \omega_i^2} \right). \tag{122}$$

In the dispersive regime, we have $|\mathrm{Im}[Z_{ii}^{(ac)}(\omega_i)]/Z_i| \ll 1$; hence, $\alpha_{ii} \simeq 1$, $|\alpha_{ij}| = |\mathrm{Im}[Z_{ij}(\omega_i)]/Z_i| \ll 1$ for $1 \leq i, j \leq N$ and $|\alpha_{i,k+N}| = |r_{ki} C_i^{1/2} (\omega_{R_k}^2/\omega_{R_k}^2 - \omega_i^2)| \ll 1$ for $1 \leq k \leq M$; hence, we can treat α_{ij} and $\alpha_{i,k+N}$ as small parameters. $\mathrm{Im}[Z_{ii}^{(ac)}(\omega_i)] = \sum_{k=1}^{M} ([A_k]_{ii} \omega_i / \omega_{R_k}^2 - \omega_i^2)$ is the ac part of $\mathrm{Im}[Z_{ii}(\omega_i)]$.

Similarly, resonator fluxes $\boldsymbol{\Phi}_R = (\Phi_{R_1}, \ldots, \Phi_{R_M})$ in the initial frame can also be related to the flux coordinates $\boldsymbol{\phi}$ in the final frame

$$\Phi_{R_k} = \sum_{i=1}^{N} \alpha_{k+N,i} \phi_i + \phi_{k+N} \tag{123}$$

where

$$\alpha_{k+N,i} = r_{ki} C_i^{1/2} \left(\frac{\omega_i^2}{\omega_i^2 - \omega_{R_k}^2} \right) \tag{124}$$

and $\alpha_{k+N,k+N} = 1$ for $1 \leq k \leq M$ by (123).

The expression for the coefficients $\beta_{pp'qq'}$ is given in [25] as

$$\beta_{pp'qq'} = \sum_{s=1}^{N} \frac{e^2}{24 L_J^{(s)}} \xi_{sp} \xi_{sp'} \xi_{sq} \xi_{sq'} \tag{125}$$

where $L_J^{(s)}$ is the inductance of the sth junction. In our case, $\xi_{sp} = \alpha_{sp}/\sqrt{\omega_p}$ after the introduction of mode operators as $\hat{\phi}_p = (\hbar Z_p/2)^{1/2}(\hat{a}_p + \hat{a}_p^\dagger)$ in the final frame with characteristic impedance $Z_p = 1/\omega_p$, one gets

$$\beta_{pp'qq'} = \sum_{s=1}^{N} \frac{E_C^{(s)}}{12} \omega_{J_s}^2 (\omega_p \omega_{p'} \omega_q \omega_{q'})^{-1/2} \alpha_{sp} \alpha_{sp'} \alpha_{sq} \alpha_{sq'} \tag{126}$$

where $E_C^{(s)} = (e^2/2C_s)$ and $\omega_{J_s} = 1/\sqrt{L_{J_s} C_s}$.

$\gamma_{pp'}$ is given in [25] as

$$\gamma_{pp'} = 6 \sum_{q=1}^{N+M} \beta_{qqpp'}. \tag{127}$$

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We observe that γ_{pp} is of order $E_C^{(s)}$. Since $E_C^{(s)}$ is already small compared with qubit frequencies, we will be only interested in the first-order expansion of $\gamma_{pp'}$ in the small parameters α_{ij} and $\alpha_{i,k+N}$. Then, we can write the diagonal entries as

$$\gamma_{ii} \simeq 6\beta_{iiii} \simeq \frac{E_C^{(i)}\omega_{J_i}^2}{2\omega_i^2}\alpha_{ii}^2. \tag{128}$$

The off-diagonal entry γ_{ij} between qubit modes i and j is

$$\begin{aligned}\gamma_{ij} &\simeq 6(\beta_{iiij}+\beta_{jjij}) \\ &\simeq \frac{E_C^{(i)}\omega_{J_i}^2}{2\omega_i}\frac{\alpha_{ii}\alpha_{ij}}{\sqrt{\omega_i\omega_j}} + \frac{E_C^{(j)}\omega_{J_j}^2}{2\omega_j}\frac{\alpha_{jj}\alpha_{ji}}{\sqrt{\omega_i\omega_j}}.\end{aligned} \tag{129}$$

The off-diagonal entry γ_{ik} between the qubit mode i and the resonator mode k is

$$\gamma_{ik} \simeq 6\beta_{iiik} \simeq \frac{E_C^{(i)}\omega_{J_i}^2}{2\omega_i}\frac{\alpha_{ii}\alpha_{i,k+N}}{\sqrt{\omega_i\omega_{R_k}}}. \tag{130}$$

In addition, the diagonal resonator entries $\gamma_{kk} \simeq 0$ to first order in α_{ij} and $\alpha_{i,k+N}$. Hence, we can write

$$\boldsymbol{\gamma} = \sqrt{\mathbf{z}}\boldsymbol{\alpha}^T\boldsymbol{\Lambda}\boldsymbol{\alpha}\sqrt{\mathbf{z}} \tag{131}$$

where $\boldsymbol{\Lambda}$ is the $(N+M)\times(N+M)$ diagonal matrix with entries $(E_C^{(i)}\omega_{J_i}^2/2\omega_i)$ for $1 \le i \le N$ and zero otherwise, that is

$$\boldsymbol{\Lambda} = \begin{pmatrix} \frac{E_C^{(1)}\omega_{J_1}^2}{2\omega_1} & & & \mathbf{0} \\ & \ddots & & \\ & & \frac{E_C^{(N)}\omega_{J_N}^2}{2\omega_N} & \\ \mathbf{0} & & & \mathbf{0}_{M\times M} \end{pmatrix} \tag{132}$$

and $\sqrt{\mathbf{z}}$ is the diagonal matrix holding the square roots of the characteristic impedances $Z_i = 1/\omega_i$ of the modes in the final frame

$$\sqrt{\mathbf{z}} = \begin{pmatrix} 1/\sqrt{\omega_1} & & \\ & \ddots & \\ & & 1/\sqrt{\omega_{N+M}} \end{pmatrix}. \tag{133}$$

H_γ in (114) can then be written as

$$H_\gamma = -\sum_{pp'}\gamma_{pp'}\left(2\hat{a}_p^\dagger a_{p'} + \hat{a}_p\hat{a}_{p'} + \hat{a}_p^\dagger\hat{a}_{p'}^\dagger\right) = -\frac{1}{2}\hat{\boldsymbol{\phi}}^T\boldsymbol{\gamma}'\hat{\boldsymbol{\phi}} \tag{134}$$

where

$$\boldsymbol{\gamma}' = \left(\frac{4}{\hbar}\right)\sqrt{\mathbf{z}}^{-1}\boldsymbol{\gamma}\sqrt{\mathbf{z}}^{-1}. \tag{135}$$

Using (131)

$$\boldsymbol{\gamma}' = \left(\frac{4}{\hbar}\right)\boldsymbol{\alpha}^T\boldsymbol{\Lambda}\boldsymbol{\alpha}. \tag{136}$$

Then, one can show that H_γ when transformed back to the original frame becomes

$$\begin{aligned} H_\gamma &= -\frac{1}{2}\boldsymbol{\phi}^T\boldsymbol{\gamma}'\boldsymbol{\phi} \\ &= -\frac{2}{\hbar}\boldsymbol{\phi}^T\boldsymbol{\alpha}^T\boldsymbol{\Lambda}\boldsymbol{\alpha}\boldsymbol{\phi} \\ &= -\frac{2}{\hbar}\boldsymbol{\Phi}^T\begin{pmatrix}\mathbf{C}_0^{1/2} & \mathbf{0}\\ \mathbf{0} & \mathbf{1}\end{pmatrix}\boldsymbol{\Lambda}\begin{pmatrix}\mathbf{C}_0^{1/2} & \mathbf{0}\\ \mathbf{0} & \mathbf{1}\end{pmatrix}\boldsymbol{\Phi} \\ &= -\frac{1}{2}\boldsymbol{\Phi}_J^T\mathbf{L}_0^{-1}\boldsymbol{\Phi}_J \end{aligned} \tag{137}$$

where $\mathbf{L}_0$ is a diagonal inductance matrix

$$\mathbf{L}_0 = \mathbf{L}_J\begin{pmatrix} \frac{\hbar\omega_1}{2E_C^{(1)}} & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \frac{\hbar\omega_N}{2E_C^{(N)}} \end{pmatrix}. \tag{138}$$

Now we write the initial Hamiltonian H_0 by adding and subtracting the term H_γ as

$$H_0 = H_0 + H_\gamma - H_\gamma = H_0' - H_\gamma \tag{139}$$

where $H_0' = H_0 + H_\gamma$. So instead of starting our treatment with H_0 if we start with an initial linear Hamiltonian H_0', we would cancel out the term H_γ that is generated by the nonlinearities. This requires an update of the junction inductances in the initial frame as follows:

$$\mathbf{L}_J^{-1} \to \mathbf{L}_J^{-1} - \mathbf{L}_0^{-1}. \tag{140}$$

That is

$$L_{J_i}^{-1} \to L_{J_i}^{-1}\left(1 - \frac{2E_C^{(i)}}{\hbar\omega_i}\right). \tag{141}$$

Hence, we can write the equation for ω_i as

$$\omega_i^2 = \omega_{J_i}^2\left(1 - \frac{2E_C^{(i)}}{\hbar\omega_i}\right) \tag{142}$$

or if we put $r = (E_C^{(i)}/\hbar\omega_{J_i})$ and $x = \omega_i/\omega_{J_i}$

$$x^2 = (1 - 2r/x). \tag{143}$$

In the limit of small anharmonicities $r \ll 1$, the solution is $x = 1 - r/(1-r)$ or

$$\omega_i = \omega_{J_i} - \frac{E_C^{(i)}/\hbar}{1 - E_C^{(i)}/(\hbar\omega_{J_i})}. \tag{144}$$

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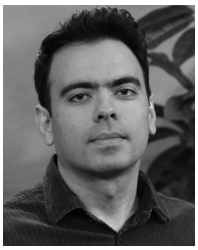
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